DEEP LEARNING-BASED MULTI-OBJECT DETECTION OF EPIPHYTES AND PARASITIC PLANTS ON HOST TREES USING OPTIMIZED MASK R-CNN FOR SYMBIOTIC RELATIONSHIP

A Thesis Presented to the Faculty, College of Computer Studies
Laguna State Polytechnic University
Los Baños Campus
Los Baños, Laguna

In Partial Fulfillment of the Requirement for the Degree
Bachelor of Science in Computer Science
Specialized in Intelligent Systems

ALEJANDREI APOLO M. DURAN
CARLOS MIGUEL Z. ROSALES

CRISANTO F. GULAY

JUNE 2026

APPROVAL SHEET

The thesis entitled "**DEEP LEARNING-BASED MULTI-OBJECT DETECTION OF EPIPHYTES AND PARASITIC PLANTS ON HOST TREES USING OPTIMIZED MASK R-CNN FOR SYMBIOTIC RELATIONSHIP**" prepared and submitted by **Alejandrei Apolo M. Duran**, and **Carlos Miguel Z. Rosales** in partial fulfillment of the requirements for the degree Bachelor of Science in Computer Science specialized in Intelligent Systems is hereby recommended for approval and acceptance.

CRISANTO F. GULAY, DIT
Adviser

Approved and Accepted by the Committee on Oral Examination with a grade of____.

CAROLINA R. JOVAL
Member

JHOCO S. MILLANES
Member

JEFFERSON L. LERIOS
Member

Accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science specialized in Intelligent Systems.

LOYD S. ECHALAR, DIT
Associate Dean, College of Computer Studies

OFELIA B. MANINGAS, EdD.
Chairperson, Research and Development Office

_____
Date Signed

RESEARCH CONTRIBUTION NO.	

ACKNOWLEDGEMENT

The authors would like to express their heartfelt gratitude to everyone who supported and guided them throughout the completion of this research work. They are deeply thankful to Almighty God for granting them the strength, wisdom, and determination to accomplish this study despite the challenges along the way.

The authors also convey their deep appreciation to Laguna State Polytechnic University for fostering a supportive learning environment that contributed to their development as students.

The authors are deeply grateful to their research adviser, Dr. Crisanto F. Gulay, whose unwavering support and insightful guidance played a crucial role in shaping this study.

They also acknowledge the University President, Dr. Mario R. Briones, and the OIC Campus Director, Engr. Manuel Luis R. Alvarez, for their leadership and for cultivating a learning environment that promotes academic excellence.

To the Associate Dean of the College of Computer Studies, Dr. Loyd S. Echalar, for creating an environment that inspires students to strive for excellence.

The authors also acknowledge the Research Implementing Unit Head, Mr. Jonardo R. Asor. Special thanks are given to the Subject Specialist, Dr. Jefferson L. Lerios, for sharing his expertise, and to the panel members, Ms. Carolina R. Joval and Mr. Jhoco S. Millanes, for their valuable recommendations that helped strengthen this research.

The authors gratefully acknowledge the Makiling Botanic Garden and their dedicated and generous staff; Analyn C Layba, Angela A. Limpiada, and Alexis Jhon T. Fajardo for granting access to their grounds and assistance during the data collection process. Their support and cooperation were instrumental in gathering the dataset for this study, and their contributions are deeply valued.

The Researchers

DEDICATION

The researchers dedicate this study wholeheartedly to Almighty God, who has been their source of strength, hope, and guidance throughout this journey. In moments when things felt overwhelming, His presence reminded them to stay patient, keep going, and trust that every effort would be worth it. This study is also dedicated to the loving parents of the researchers, for their unconditional love, constant support, and countless sacrifices. Your belief gave them the drive to push forward and finish what they started.

Mrs. Ma. Amor M. Duran

Mr. Cesar Z. Rosales

Mrs. Rosita Z. Rosales

The researchers further extend their appreciation to their siblings. Thank you for your quiet encouragement, understanding, and support along the way. To friends, thank you for standing by their side, cheering them on, and making the journey lighter with your words, laughter, and presence. This study is not just the result of their hard work, but also the outcome of the support and belief of the people who stood by them.

A.A.M.D

C.M.Z.R

ABSTRACT

Symbiotic interactions between trees and associated plant organisms, such as epiphytes and parasitic species play a crucial role in forest ecosystem, health, and biodiversity. Monitoring these organisms, however, remains difficult due to limited automated tools and the complexity of natural environments. This study implements and optimizes a Mask R-CNN instance-segmentation model to detect and segment epiphyte plants, parasitic plants, and host trees in forest settings. A total of 1,661 augmented images were used, and the model was trained initially with a ResNet-50 backbone before introducing improvements through a deeper ResNet-101. Model performance was evaluated using mAP@0.5 IoU. The baseline model achieved 73.088% mAP@0.5 for bounding boxes and 67.694% for segmentation masks. After completing the baseline training, the study identified several performance limitations and challenges, then applied the appropriate optimization techniques to address them. Following optimization, the model reached 83.007% for bounding boxes and 76.746% for segmentation, reflecting performance increases of 13.57% and 13.37%, respectively. The optimized model demonstrates stronger detection of small, partially occluded epiphytes and produces cleaner and more consistent masks for parasitic plants and host trees. These results confirm the potential of Mask R-CNN as a practical tool for automated biodiversity monitoring and provide a strong baseline for future research focused on forest health assessment, long-term ecological studies, and the development of advanced plant–tree interaction monitoring systems.

Keywords: *Symbiotic Interaction, Mask R-CNN, Instance-Segmentation Model, Detect, Segment, Epiphyte Plants, Parasitic Plants, Host Trees, Restnet-50 Backbone, Restnet-101, maAP0.5, Bounding Box, Segmentation Masks*

TABLE OF CONTENTS

Preliminaries	***Page***
Title Page	i
Approval Sheet	ii
Acknowledgement	iii
Dedication	v
Abstract	vi
Table of Contents	vii
List of Figures	ix
List of Tables	xi
Definition of Terms	xiii

CHAPTER

I	**INTRODUCTION**	
	Background of the Study	1
	Objectives of the Study	5
	Significance of the Study	5
	Scope and Limitations of the Study	7
II	**REVIEW OF RELATED LITERATURE AND STUDIES**	
	Review of Related Literature, Studies, and Systems	11
	Synthesis and Relevance of the Study	25
	Related Literature Matrix	27
	Conceptual Framework	30
III	**METHODOLOGY**	
	Research Design	33
	Study Area	33
	Hardware	35
	Software	35
	Frameworks and Libraries	37
	Data Collection Method	40
	Model Development	40
	Data Acquisition	41
	Model Training	45
	Model Evaluation	46
	Model Optimization	48
	Experimental Setup	49
	Ethical Consideration	51
IV	**RESULTS AND DISCUSSIONS**	
	Initial Training Setup for Mask R-CNN	52
	Result of the Initial Implementation of Mask R-CNN	54
	Performance Overview and Identification of Challenges	59
	Training Setup for Optimization of Mask R-CNN	61
	Result of the Implementation of Optimized Mask R-CNN	64
	Statistical Test	69

V SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

Summary 73
Conclusions 74
Recommendations 75

REFERENCES
APPENDICES

Appendix A. Approval Forms
Appendix B. Certificate of Acceptance
Appendix C. Relevant Source Code
Appendix D. Evaluation Tool
Appendix E. Sample Input / Output
Appendix F. Curriculum Vitae
Appendix G. Matrix for Revisions
Appendix H. Grammarly Report
Appendix I. Turnitin Report
Appendix J. Publishable Paper (IEEE FORMAT)

LIST OF FIGURES

Figure	Title	Page
1	Conceptual Framework for Deep Learning-Based Multi-Object Detection of Epiphytes and Parasitic Plants on Host Trees Using Optimized Mask R-CNN for Symbiotic Relationship	32
2	Model Development Diagram	41
3	Experimental Setup Diagram	49
4	Training Loss of Best Result in Initial Implementation of Mask R-CNN	57
5	Validation mAP Plot of Best Result in Initial Implementation of Mask R-CNN	58
6	Confusion Matrix for Bounding Box Level of Best Result in Initial Implementation of Mask R-CNN	58
7	Confusion Matrix for the Segmentation Level of Best Result in Initial Mask R-CNN	59
8	Overview of Object Detection and Segmentation of the Initial Output of Epiphyte Plants on Host Trees	60
9	Overview of Object Detection and Segmentation of the Initial Output of Parasitic Plants on Host Trees	60
10	Training Loss Plot of Best Result in Optimized Mask R-CNN	66
11	Validation mAP Plot of Best Result in Optimized Mask R-CNN	67
12	Confusion Matrix for bounding box level of Best Result in Optimized Mask R-CNN	67

13	Confusion Matrix for the segmentation level of Best Result in Optimized Mask R-CNN	68
14	Overview of Object Detection and Segmentation of Optimized Output of Epiphyte Plants on Host Trees	68
15	Overview of Object Detection and Segmentation of Optimized Output of Epiphyte Plants on Host Trees	69

.

LIST OF TABLES

Table	Title	Page
1	Matrix of Reviewed Related Literature	29
2	Raw Image Dataset Collected	41
3	Total Number of Annotated, Augmented, and Preprocessed Images	44
4	Total Instances of Classes	45
5	Total Number of Annotated, Augmented, and Preprocessed Initial Image Datasets	53
6	Total Number of Instances per Class of Initial Datasets	53
7	Details of Augmentations Used in Initial Model Training	54
8	Details of Hyperparameters Used in Initial Model Training	54
9	Results from the Initial Implementation in Mask R-CNN	56
10	Results of the Best Performing Initial Implementation of Mask R-CNN	57
11	Total Number of Cleaned and Optimized Datasets	62
12	Total Number of Instances per Class of Cleaned and Optimized Datasets	63
13	Details of Preprocessing Used in Optimized Model Training	63
14	Details of Augmentations Used in Optimized Model Training	63
15	Details of Hyperparameters Used in Optimized Model Training	64
16	Results from the Optimized Implementation in Mask R-CNN	65
17	Results of the Best Performing Optimized Implementation of Mask R-CNN	66

18	Descriptive Statistics for Both Expert Evaluation and Optimized Mask R-CNN Prediction	70
19	Result of Paired Samples T-Test for Both Expert Evaluation and Optimized Mask R-CNN Prediction	72

DEFINITION OF TERMS

Backbone Network	The part of a deep learning model that extracts features from an image. In this study, ResNet-50 is used as the initial backbone, and ResNet-101 is considered for optimization.
Bounding Box	A rectangular box used to indicate the location of an object in an image. It helps with labeling and detecting objects like epiphytes or parasitic plants.
Confusion Matrix	A table used to show the model's correct and incorrect predictions for each class.
Data Augmentation	Techniques used to artificially increase the size and variety of the dataset. In this study, it includes flipping, rotation, and brightness adjustments to improve model performance especially to double the number of dataset size
Data Preprocessing	Preparing raw images for model training. This includes resizing, cleaning, and formatting images to make them suitable for learning.
Data Splitting	Dividing the dataset into training, validation, and testing sets. This helps evaluate the model's ability to generalize new images.
Dataset	A collection of images used for training and testing the model. In this study, the dataset contains labeled images of epiphytes, parasitic plants, and host trees.
Deep Learning	A type of machine learning that uses layered neural networks to process data. This study uses deep learning to train a model that can detect plant symbiotic interactions in images.
Detectron2	A deep learning framework used in this study to implement the Mask R-CNN model.

Epiphyte Plants	Plants that grow on the surface of other trees but do not take nutrients from them. In this study, epiphytes represent the commensalism type of symbiotic relationship.
Evaluation Metrics	Standard measurements used to assess the performance of the model.
F1-score	The balance between precision and recall.
Google Colab	An online platform used in this study to train and test the model using cloud-based computing resources.
Host Tree	The tree on which either epiphytes or parasitic plants grow. It serves as the base object in the image for detecting plant symbiotic interactions.
Hyperparameter Tuning	The process of adjusting settings like learning rate or batch size to improve model performance.
Image Annotation	The process of labeling objects in an image to train machine learning models. This study involves marking epiphytes, parasitic plants, and host trees using instance masks.
Instance Mask	A pixel-level outline of a detected object. In this study, it is used to provide a detailed shape of each plant object detected by the Mask R-CNN model.
Instance Segmentation	A technique in computer vision that identifies and outlines each object separately in an image. In this study, it is used to locate and separate plant objects from the background.
IoU (Intersection over Union)	Measures how much the predicted object overlaps with the actual object.
Manual Data Collection	The process of capturing images personally instead of downloading from the internet. In this study, images were manually collected from real forest or green areas.

mAP (mean Average Precision)	Measures the overall detection accuracy across all classes.
Mask R-CNN	A deep learning model used for object detection and instance segmentation. It is applied in this study to detect and segment epiphytes, parasitic plants, and host trees in natural outdoor images.
Object Detection	A computer vision task that involves identifying and locating objects within an image. This study applies object detection to identify plant components in tree images.
Overfitting	When the model performs well on training data but poorly on new or unseen data. The study monitors this during validation.
Parasitic Plants	Plants that attach to host trees and extract water and nutrients, which can harm the host. In this study, they represent the parasitism type of symbiosis.
Precision	The percentage of correct positive predictions out of all predicted positives.
Python	The programming language used to write and run the model code in this research.
Recall	The percentage of correct positive predictions out of all actual positives.
ResNet (Residual Network)	A type of neural network used as a backbone in object detection models. It helps improve accuracy by handling deep layers effectively.
Roboflow	A tool used in this study for image annotation and dataset preparation.
Symbiotic Relationship	A close interaction between different plant species. This study focuses on two types: commensalism (epiphytes) and parasitism (parasitic plants).

CHAPTER I

INTRODUCTION

BACKGROUND OF THE STUDY

Symbiosis refers to a close and long-term interaction between two different species, which can be either beneficial, neutral, or harmful to one or both partners (Singh & Kumar, 2025; Smith & Read, 2008). Mutualism is an interaction between two species that benefits both of them (Bronstein, 2009). Commensalism is a relationship between two species in which one species benefits while the other species does not experience any harm or benefit (Mathis & Bronstein, 2020). One kind of symbiotic interaction known as parasitism occurs when one species gains an advantage over the other species while causing it harm (Alhadi et al., 2024). In many forested areas, plants rely on trees as hosts, creating diverse ecological interactions that directly affect the health and sustainability of the environment (Bottan et al., 2025). In forest ecosystems, these relationships are essential for nutrient cycling, seed dispersal, and even disease resistance (Van Der Heijden et al., 2015).

Mycorrhizal fungi form mutualism relationship with tree roots to help absorb water and nutrients while receiving sugars in return (Ibáñez et al., 2025). Parasitic plants harm trees by using specialized structures to steal water, nutrients, and sugars from their tissues, weakening or even damaging the host, which gains no benefit in return (Ali, 2025). Epiphyte plants form commensalism relationship that lives on the surface of trees without harming them but still depend on the host for support and access to sunlight (Zotz, 2013). These interactions are complex and highly dynamic, making them crucial for

understanding how ecosystems adapt to changing environments (Rodriguez et al., 2009).

About 10% of all vascular plant species are epiphytes, showing their significant presence in forest ecosystems (Zotz & Hietz, 2001). In tropical rainforests, epiphyte plants can make up as much as 50% of the plant biodiversity in the canopy layer (Nadkarni, 1994). Moreover, mycorrhizal fungi are associated with over 80% of all terrestrial plant species, indicating how widespread these symbiotic relationships are across different environments (Smith & Read, 2008). In one study found that nearly 70% of the tree species examined hosted at least one kind of epiphyte plant or parasitic plant (Gentry & Dodson, 1987). These numbers suggest that detecting and monitoring these organisms is vital for tracking overall forest health and biodiversity.

In Costa Rica, studies have shown that epiphyte plants play a significant role in maintaining moisture levels in cloud forests, acting like sponges that trap rainwater and fog (Nadkarni & Matelson, 1992). In Southeast Asia, parasitic plants like *Rafflesia* have evolved unique relationships with their hosts, entirely relying on them for nutrients without photosynthesizing (Barkman et al., 2004). Similarly, in African forests, mistletoes are known to parasitize a wide variety of tree species, impacting their growth and survival (Watson, 2001). In Europe, lichens, often considered a form of mutualistic symbiosis between fungi and algae, are commonly used as indicators of air quality (Nimis et al., 2002). These global examples highlight the diversity of symbiotic interactions and their environmental significance across regions.

In the Philippines, forests host a wide variety of epiphyte plants, including orchids, ferns, and mosses, many of which are endemic and endangered (Pelser et al., 2011). Some Philippine trees also support parasitic plants like *Loranthus* and *Dendrophthoe*, which attach to branches and extract nutrients from the host (Barcelona et al., 2006). These interactions often go unnoticed due to the difficulty of monitoring them in dense forest canopies, especially in remote regions like the Sierra Madre and Mindanao (Cruz et al., 2019). Moreover, traditional ecological studies in the Philippines lack large-scale data on symbiotic relationships due to limited technological and resource availability (Ramos & Malabrigo, 2020). While international studies have successfully adopted remote sensing for mangrove monitoring (Pillodar et al., 2023), biodiversity monitoring in the Philippines remains largely dependent on manual surveys, which are slow and labor-intensive.

One major problem is the increasing threat of habitat loss, which disrupts existing symbiotic relationships and prevents new ones from forming (Turner et al., 2003). Additionally, invasive species can replace or compete with native epiphyte plants and parasitic plants, altering the natural balance within ecosystems (Mack et al., 2000). Climate change also impacts the timing and distribution of symbiotic interactions, especially for species that rely on very specific environmental conditions (Rodriguez et al., 2009). Another issue is the lack of consistent monitoring tools, which makes it difficult to detect changes in the abundance or health of these organisms (Zhao et al., 2019). These challenges create gaps in ecological data and reduce the effectiveness of conservation strategies. Study in Panay Island revealed a 20% decline in mangrove cover over three decades, highlighting both the urgency of

monitoring and the insufficiency of current approaches in capturing ecosystem complexity (Garrido et al., 2021).

One effective solution is the integration of deep learning and computer vision to automate the identification of symbiotic organisms in field images (Ren et al., 2015). With object detection models like Mask R-CNN, researchers can classify and segment epiphyte plants and parasitic plants from their host trees more efficiently (He et al., 2017). The original Mask R-CNN framework introduced high-precision object detection, making it well-suited for analyzing complex forest environments. In ecological applications, Mask R-CNN combined with object-based image analysis improved segmentation accuracy by 25% in dryland vegetation (Zhou et al., 2021). Similarly, UAV-based applications for plant counting and sizing have demonstrated its effectiveness in handling complex canopy structures (Li et al., 2020). Extensions of Mask R-CNN have also been developed for phenotypic estimation in crops (Yu et al., 2024) and nested object detection, such as bees and parasites (Wang et al., 2024), proving its adaptability for detecting small and dependent organisms. Moreover, optimizing the hyperparameters of these models enhances accuracy and improves adaptability to various forest types (Li et al., 2020).

Given these developments, there is an immediate requirement to establish an optimized Mask R-CNN framework specifically designed for the automated detection and segmentation of epiphyte plants and parasitic plants on host trees in forests. Unlike existing ecological monitoring approaches that rely on manual observation or broad-scale satellite imaging, this study directly addressed the gap in automated biodiversity monitoring and served as a

baseline by providing a reliable and accurate model solution for detecting and segmenting plants and host trees involved in symbiotic interactions. Through this innovation, the study contributes to both technological developments and practical conservation efforts by enhancing the understanding, monitoring, and preservation of symbiotic relationships within forest ecosystems.

OBJECTIVES OF THE STUDY

The main objective of this study is to implement and optimize a Mask R-CNN-based deep learning model for accurate detection and segmentation of epiphyte plants, parasitic plants, and their host trees in natural outdoor images. To further achieve this, it is necessary to initially accomplish the following objectives:

1. To implement the Mask R-CNN model for multiple object detection and segmentation of epiphyte plants, parasitic plants, and host trees using still image data related to plant symbiosis.
2. To improve the detection and segmentation performance of a Mask R-CNN model for epiphyte plants and parasitic plants with their host trees by applying optimized data preparation and model tuning strategies.
3. To evaluate the performance of the Mask R-CNN model under real-world conditions using standard object detection and segmentation metrics within the study period.

SIGNIFICANCE OF THE STUDY

This study explores the use of deep learning, specifically Mask R-CNN, for detecting multiple plant-related objects, such as epiphytes, parasitic plants, and host trees, in a single image captured from real outdoor environments.

These organisms are known for forming symbiotic relationships that can influence the health of trees and the surrounding ecosystem. However, monitoring these interactions in natural settings remains limited due to the lack of effective and practical approaches. By using deep learning-based object detection, this research presents a model that can support the observation and assessment of these organisms in complex forest environments.

The study provides a simplified method of detection by using general object categories instead of requiring species-level classification, making it more adaptable to real-world applications. It works with manually captured images under uncontrolled conditions, such as natural lighting and varied tree structures, giving the study realistic value. By optimizing the model based on detection performance, the research helps address key challenges in instance segmentation, especially in cluttered and dynamic backgrounds. The output of this study contributes to the field of computer vision by applying deep learning techniques in a non-industrial, environmentally focused domain.

This work also aligns with several of the United Nations Sustainable Development Goals (SDGs). It supports SDG 15, Life on Land, which focuses on protecting, restoring, and promoting the sustainable use of terrestrial ecosystems and halting biodiversity loss. It also contributes to SDG 13, Climate Action, by encouraging the use of technology in monitoring changes in forest health that may be related to climate impacts. Additionally, it supports SDG 9, Industry, Innovation and Infrastructure, by applying advanced deep learning methods in the field of environmental monitoring.

UNESCO (2022) has highlighted the importance of using artificial intelligence in biodiversity tracking and ecological research. By implementing a

deep learning-based object detection approach in the study of plant symbiosis, this research contributes to global efforts in building innovative and sustainable solutions for ecosystem monitoring and land management. Specifically, the study would be beneficial to the following:

Environmental and Agricultural Researchers: It provides a deep learning-based framework that may support further studies in vegetation monitoring, symbiotic interaction analysis, and forest data collection.

Forestry Officers and Conservation Planners: The detection of parasitic plants and epiphytes can help monitor host tree conditions and assess ecological balance in managed or protected areas.

Computer Science Students and Future Researchers: The study demonstrates how models like Mask R-CNN can be used in non-traditional fields, such as environmental science, offering real-world applications of deep learning in plant monitoring.

Sustainability and Climate Advocates: The model presented in this study may serve as a reference for integrating artificial intelligence into conservation and climate response strategies, promoting smarter and data-driven environmental solutions.

SCOPE AND LIMITATIONS

This study focuses on implementing a deep learning model, specifically Mask R-CNN, for multiple object detection of epiphytes and parasitic plants on their host trees. These plant organisms are commonly found in symbiotic relationships, where one benefits while the other may or may not be affected. The goal is to detect and classify them using image data, labeling each object as either an epiphyte, a parasitic plant, or a host tree. The study does not aim

to identify specific species, but only their general categories, since epiphytes already represent commensalism, and parasitic plants represent parasitism symbiotic relationship.

The main objective of this study is to evaluate how accurately Mask R-CNN can detect, segment, and classify these plant organisms and the host trees in a single image. The focus is on detection accuracy rather than processing speed or real-time performance. All tests are done using still images, meaning the study does not involve videos or live footage. This setup allows for a controlled testing environment, free from complications such as movement, lighting changes, or frame rate issues. After the initial implementation, possible detection errors such as misclassification, incorrect segmentation, or missed objects were identified. These issues guided the optimization process, which may involve tuning hyperparameters, optimizing the backbone network, improving image preprocessing, or expanding the training data through augmentation.

This research aims to support the use of deep learning algorithms in solving real-world challenges in agriculture, forest monitoring, and environmental conservation. By using a deep learning-based approach to detect epiphytes and parasitic plants on host trees, the study offers a practical method for observing plant interactions that influence forest and ecosystem health. The results may be useful for environmental researchers, forestry officers, agricultural field technicians, conservation planners, and research-based organizations that monitor vegetation and tree conditions. Through better data collection and analysis, the study can help improve fieldwork

efficiency, support sustainable land use, and contribute to long-term efforts in maintaining healthy and balanced ecosystems.

However, this study has several limitations. First, it focuses only on image-based detection using static photos. It does not test the model in real-time applications such as drone footage, mobile apps, or live video streams. As a result, the model's performance in dynamic or time-sensitive environments is not assessed. Second, detecting epiphytes and parasitic plants on trees does not mean that both are present on the same tree, due to the limited availability of datasets that show such co-occurrence.

Another challenge is the complexity of natural outdoor environments captured in the images. Since these organisms can appear in different shapes, sizes, and positions, and may sometimes be partially hidden or blend into the tree structure, the model may face difficulty in consistently segmenting and identifying each object. These factors can affect detection accuracy and may lead to overlapping object boundaries or false classifications.

The quality of the dataset is also a critical factor. In this study, the dataset is primarily built through manual photography due to the lack of publicly available, high-quality images that fit the scope. The images were captured in accessible forested or green areas where these plant interactions naturally occur. While this helps reflect real-world conditions, such as natural lighting, different structures of the trees, and how the plants naturally grow on trees, it also introduces challenges like uneven lighting, blur, or inconsistent object visibility. All images are manually annotated to identify the target classes, which are the epiphytes, parasitic plants, and host trees. The dataset size is kept manageable to balance training quality and processing time.

To measure the model's performance, standard object detection metrics are used, including precision, recall, mean Average Precision (mAP), F1-score, Intersection over Union (IoU), and the confusion matrix. These metrics help evaluate how accurately the model detects and classifies each object in the image and also guide optimization by identifying the areas where the model struggles most.

In conclusion, this study is focused on implementing and optimizing Mask R-CNN for detecting epiphytes, parasitic plants, and their host trees using still images taken in natural outdoor settings. The research is limited to image-based detection and does not include real-time or video applications. It also does not aim to identify plant species or detect both plant types on the same tree. Since these organisms vary in appearance and are found in visually complex environments, the model may still face challenges. Despite these limitations, the study aims to produce results that can support agriculture, forest health, biodiversity research, and conservation. After the initial testing, any issues found in detection were analyzed to improve the model's performance and reliability.

CHAPTER II

THEORETICAL FRAMEWORK OF THE STUDY

REVIEW OF RELATED LITERATURE, STUDIES AND SYSTEMS

This chapter provides a comprehensive review of literature, studies, and systems that is relevant to the symbiotic relationship of plant to tree host. By examining existing knowledge, it gathered key findings that served as significant information, identified gaps, and highlighted challenges encompassing issues that needed to be addressed. Specifically, it focuses on exploring different approaches, algorithms, techniques, and methods which can be used in the development.

Significance of Symbiotic Relationship Detection

Understanding the presence and role of symbiotic organisms on trees, such as epiphytes and parasitic plants, is vital for monitoring forest health and biodiversity (Smith & Read, 2008). These relationships can be either beneficial, neutral, or harmful that can influence nutrient cycles, forest growth, and overall ecosystem stability (Van Der Heijden et al., 2015). Forest is increasingly affected by human activities and climate change, detecting such these interactions can serve as an early warning system for environmental observation (Rodriguez et al., 2009). Moreover, detecting epiphytes and parasitic plants can help forest managers and researchers make better decisions about conservation and restoration (Zotz, 2013). Therefore, integrating deep learning into this detection process not only improves accuracy but also supports ecological sustainability efforts in a more efficient way (Krizhevsky et al., 2012).

Additionally, epiphytes often serve as bioindicators of air quality and climate conditions due to their sensitivity to environmental changes (Nadkarni, 1994). Unlike parasitic plants that negatively impact their hosts, epiphytes typically co-exist without causing harm, making them useful indicators of healthy forest systems (Zotz & Hietz, 2001). However, both types of symbiotic organisms need to be accurately identified and monitored because their abundance and behavior can reflect broader ecosystem trends (Gentry & Dodson, 1987). Unfortunately, traditional manual monitoring methods are labor-intensive and prone to human error, especially in dense or remote forest areas (Turner et al., 2003). Therefore, the application of deep learning models offers a scalable and efficient solution for long-term observation and ecological analysis (Ren et al., 2015).

In recent years, image-based detection systems have gained popularity in forest ecology due to their efficiency and ability to process large datasets (Zhou et al., 2020). Integrating object detection algorithms like Mask R-CNN allows for pixel-level segmentation, which is especially useful when distinguishing between overlapping plant species (He et al., 2017). This is particularly relevant for epiphytes and parasitic plants, which often grow close together on tree surfaces and can be hard to tell apart in regular RGB images (Barthlott et al., 2001). Therefore, the automation of symbiotic relationship detection is not just a technical achievement but also a tool for advancing ecological research and forest policy development (Bruna & Ribeiro, 2005). With the increasing availability of labeled image datasets, such AI-powered systems are becoming more accurate and accessible for environmental applications (Lin et al., 2014).

Challenges of Manual and Traditional Identification of Symbiotic Relationship

Traditional identification of epiphytes and parasitic plants on trees has required field observation and manual annotation by trained ecologists, which is time-consuming and prone to human error (Turner et al., 2003). In dense or remote forest environments, collecting accurate data becomes even more difficult due to limited visibility, seasonal variations, and the diversity of plant species present (Zotz, 2013). Moreover, physical access to the upper canopy where many epiphytes grow is not always possible, making it hard to get a complete view of the symbiotic relationships in a given area (Nadkarni, 1994). Although preserved plants and manual classification offer historical references, they often lack real-time updates and cannot efficiently reflect dynamic ecological changes (Rodriguez et al., 2009). Therefore, manual methods alone are not sufficient to support large-scale or long-term monitoring efforts for understanding tree-organism interactions (Barthlott et al., 2001).

Furthermore, the visual similarities between epiphytes and parasitic plants often lead to misidentification during manual surveys, especially when dealing with overlapping plant structures or similar leaves(Zotz & Hietz, 2001). Some symbiotic species also mimic the appearance of their host or surrounding plants, further complicating their identification without advanced tools (Gentry & Dodson, 1987). Due to these challenges, traditional techniques may not provide consistent accuracy across different forest conditions and biodiversity levels (Bruna & Ribeiro, 2005). Furthermore, relying solely on human expertise can introduce bias depending on the researcher’s familiarity with specific plant species (Smith & Read, 2008). As a result, researchers have started to explore

automated image-based solutions to improve the efficiency and accuracy of species identification (Ren et al., 2015).

Object Detection

Object detection plays an essential role in computer vision as it enables the precise identification and localization of multiple objects within an image or video (Jiao et al., 2019). Consequently, the field has advanced significantly with the adoption of deep learning, allowing for better generalization across different environments (Beery et al., 2018). Moreover, the two main strategies for object detection, namely the one-stage and two-stage approaches, offer different trade-offs between speed and accuracy (Du et al., 2020). In addition, one-stage detectors like YOLO are favored for real-time applications due to their faster inference times (Redmon & Farhadi, 2018). However, two-stage detectors such as Faster R-CNN continue to be preferred for tasks requiring higher accuracy despite their increased computational cost (Ren et al., 2015).

Object detection techniques are continuously enhanced by introducing innovative feature extraction methods and backbone architectures (Hossain et al., 2019). For instance, deep convolutional neural networks (CNNs) have significantly improved the ability to detect objects across various scales and complexities (Qian et al., 2018). Additionally, the applications of object detection span across critical areas such as autonomous driving, medical imaging, and agricultural monitoring (Jalled & Voronkov, 2016). Specifically, deep learning-powered object detectors have been widely adopted for pest detection in crops, improving agricultural productivity (Liu et al., 2022). Overall,

the integration of object detection technologies has reshaped numerous industries by enhancing automation and decision-making processes.

Object detection aims to locate all the occurrences of the objects, whether in an image or video, regardless of their class (Sharma & Thakur, 2017). Each detected object is said to have pose information such as its location, scale, bounding box, or segmentation mask (Amit et al., 2021). This is also a technique with the application of the neural network for detecting and classifying objects within an image (Murel & Kavlakoglu, 2024). Additionally, object detection comes from two approaches: one-stage detectors and two-stage detectors. One-stage detectors or regression-based models consist of one neural network, which is very effective and efficient in terms of detection speed for predicting or detecting bounding boxes and classes from an image (Zhang et al., 2021). However, if the study is more focus on accuracy then two-stage detectors is more suitable as it has more robustness and accuracy than one-stage detectors, as they consist of two stages, the region proposal generation and object localization, but the downside of it is that it's slower as it requires extra computational complexity and training steps (Varghese & Sambath, 2023). Furthermore, object detection is also extremely crucial in computer and robotics vision (Vahab et al., 2019), and is being applied to various fields such as in Unmanned Aerial Vehicle or UAV for security (Jalled & Voronkov, 2016), Pest Detection for solving the problem of crop losses caused by pest (Chen et al., 2021), medical image analysis for early detection and diagnosis of disease (Ganatra, 2021), application in self-driving car or autonomous driving system using the hybrid approach in object detection (Khan et al., 2023), an android based mobile phone to help visually impaired that

detect and recognize objects and humans within their surrounding which can help them for their safety (Aralikatti et al. 2020), and many more.

Deep Learning-Based Model for Multi-Object Image Detection

Deep learning-based object detection is an absolutely crucial and important branch in computer vision, as it uses artificial neural networks, particularly Convolutional Neural Network (CNN), to identify and locate objects in an image and video. Applying deep learning to object detection has indeed enhanced the performance of object detection in terms of its accuracy and speed, making it more useful and helpful in various fields (Hebbar & Pullela, 2023). This deep learning-based model is divided into categories, which are the one-stage algorithms, two-stage algorithms, and end-to-end algorithms (Xing et al., 2023). Based on this study (Zhang & Cloutier, 2022), it suggests the following aspects for better improvement of the one-stage detectors; the use of more effective pyramid structure for feature extraction, the introduction of a deeper and wider network agent architecture, enhancements to the anchor-free mechanism, and the implementation of a more robust image enhancement strategy. One of the most used one-stage detector algorithms for this is the You Only Look Once (YOLO), which consists of only one neural network that evaluates the full image in one evaluation for predicting bounding boxes and class probabilities (Redmon et al., 2016). On the other hand, the Region Proposal-based Architectures or two-stage detector indeed outperforms the one-stage detector in terms of accuracy, and additionally, in this study, it also demonstrates that there is no chance for achieving both accuracy and speed at the same time; it needs to sacrifice one to get the other (Saddique et al., 2020).

Deep learning-based object detection methods have revolutionized the way multi-object detection is performed in complex visual scenes (Garcia-Garcia et al., 2017). It is evident that CNNs are at the core of this transformation, providing the backbone for detecting multiple instances of objects with high accuracy (Dyrmann et al., 2016). In addition, improvements such as the use of transformer-based detectors and anchor-free methods have further elevated the performance of one-stage models (Carion et al., 2020). Meanwhile, popular two-stage detectors like Mask R-CNN have demonstrated superior performance in tasks that demand precise segmentation and classification (Delight & Karunakaran, 2021). Nevertheless, a persistent challenge in deep learning-based detection models remains the balance between achieving high accuracy and maintaining fast inference times (Zhu et al., 2019).

Recent advancements in one-stage detectors, particularly the improvements introduced in the YOLO series, have demonstrated significant progress in achieving both higher detection accuracy and real-time performance (Redmon & Farhadi, 2018). Although end-to-end object detectors offer simplicity and efficiency by directly predicting final object detections without intermediate steps, they often require larger datasets to reach competitive performance (Shorten & Khoshgoftaar, 2019). Therefore, strategies such as data augmentation, transfer learning, and feature pyramid networks have been utilized to address the limitations of deep learning models in multi-object detection scenarios (Kussul et al., 2017). On another note, Mask R-CNN's ability to perform instance segmentation in addition to bounding box prediction makes it particularly valuable for detailed analysis, such as identifying symbiotic relationships among organisms (Afzaal et al., 2021).

Collectively, deep learning-based models are indispensable tools for advancing multi-object detection tasks in both natural and artificial environments.

Choosing Between One-Stage and Two-Stage Detectors

Since this study primarily focuses on the accuracy over speed of the deep learning-based model in multiple-object detection, it is absolutely important to compare the one-stage and two-stage detectors. In this study (Bi et al., 2023), it states that since one-stage detectors are lightweight and can unify their structure into one stage, it suggests that it is more suited for the field with limited computational capacity, and two-stage on the other hand, is suited for fields where accuracy is more priority. Even with the fact that one-stage algorithms are faster, it is, however, vulnerable to false detection for small objects as they only consist of a small amount of pixels for each, and may not be effective to learn all the features of different categories, especially dealing with imbalanced data (Zhang, 2024). The typical two-stage detectors, which are R-CNN, FAST R-CNN, FASTER R-CNN, R-FCN, and SPP-Net are compared with one-stage detectors, which are YOLO, YOLO9000, and SSD in this study (Qian et al., 2018), with different backbones each but the same dataset of PASCAL VOC 2007, and in results, FASTER R-CNN with the backbone of ResNet-101 got the highest mAP percentage of 79.3%. Furthermore, this study (Aslam et al., 2020) suggests that even one-stage detectors can outperform two-stage detectors based on the modification of the backbone of the model. This study compares the SSD and Mask R-CNN with the same backbone of ResNet50 and the same dataset of COCO and PASCAL, and concludes that SSD with ResNet50 got the higher mAP on the full dataset for both COCO and PASCAL. Another study about using a different backbone for each algorithm

does absolutely have an effect on its performance, as this study shows that the different one-stage and two-stage detectors with different backbones achieve better results (Jiao et al., 2019).

Mask R-CNN for Multiple-Object Detection

Mask R-CNN is also a two-stage detector and is an extension of Faster R-CNN that adds a parallel branch for pixel-level object masks, allowing it to detect multiple objects and create accurate segmentation masks at the same time (He et al., 2017). Mask R-CNN works by first detecting objects in an image using bounding boxes and class labels, then generating a pixel-level mask with various RCNN techniques for each object to achieve accurate instance segmentation (Delight & Karunakaran, 2021). One of the requirements for accurate object detection is to produce a large amount of dataset, but in this study (Siricharoen et al., 2023), it shows that Mask R-CNN can achieve high accuracy even with a small dataset by using multi-object sampling techniques and optimal thresholding to improve robustness and performance. Furthermore, one of the key factors in enhancing the performance of Mask R-CNN is applying effective data augmentation and hyperparameter tuning to achieve strong mAP in complex environments (Afzaal et al., 2021). Additionally, for further enhancement of the model, this study (Wu et al., 2020) proposes an improved object detection method, RGC Mask R-CNN, which enhances standard Mask R-CNN by using ResNeXt-101-64×4d as the backbone, incorporating group normalization to address small batch size issues, and applying a three-stage cascade IoU threshold strategy to balance sample quality and prevent overfitting, resulting in superior performance on COCO and PASCAL VOC07 datasets.

In this study, (Musyarofah et al., 2020), Mask R-CNN was successfully applied for roof detection in aerial images, achieving a high accuracy of mAP of 90.14% and precision of 88.71% by using RGB features, although the segmentation masks were not perfectly aligned with the objects. This other study is somehow related again to aerial images using Mask R-CNN (Hou & Li, 2024), where the method was applied specifically for building feature extraction, achieving around 95% classification accuracy and outperforming models like CNN, R-CNN, and GAN while maintaining strong stability and generalization. Aside from building detection, Mask R-CNN-based methods have also been used for agricultural purposes, where an improved model called MASU R-CNN (Tian et al., 2020) achieved a high precision of 6.43%, and strong segmentation performance in detecting apple flowers at different growth stages. Similarly, Mask R-CNN has been applied in agriculture for disease detection, where a study introduced a strawberry disease dataset and used Mask R-CNN with a ResNet backbone to achieve an instance segmentation performance with a mean average precision of 82.43% under complex backgrounds (Afzaal et al., 2021). In another related study, Mask R-CNN was used for fish segmentation in sonar images, where the addition of a PreCNN preprocessing network helped standardize feature maps and improved the segmentation accuracy across different fish farm environments (Chang et al., 2021). Furthermore, in the field of facial recognition, Mask R-CNN has also been extended into a more specialized model called G-Mask (Lin et al., 2020), which combined face detection and segmentation in a single framework using ResNet-101, RoIAlign, and GIoU loss, resulting in improved accuracy over Faster R-CNN and Mask R-CNN on benchmarks like FDDB, AFW, and WIDER FACE.

For better object detection and segmentation accuracy of the model, it indeed requires a lot and large amount of dataset for training. Mask R-CNN requires a diverse dataset for better detection and segmentation of objects, but in this study (da Cruz et al., 2025), which only has 525 images but expanded to 2,408 images through data augmentation for their study of the detection of foliar disease in soybeans using ResNet101 as backbone network for Mask R-CNN. ResNet101 is known as a deep network that requires a large amount of dataset, however, in this study (Zikra et al., 2023), they still managed to get a good high mAP even having only 651 images as datasets and with a total of 8,922 annotated objects. Additionally, in this study (Zhou 2020), it was also stated that the issue with a limited dataset indeed affects the performance of the model, so in his study, he proposes a different augmentation technique, which is slot-based image augmentation, to strengthen the dataset with extra foreground and background combinations. Furthermore, in this study (Islam et al., 2024), they only use 1,000 image datasets trained on ResNet101, but this time it is with CB-Net, and uses transfer learning for reducing the processing time. The transfer learning is also used in this study (Chowdhury et al., 2024), for reducing processing time and for smaller datasets, although it uses various techniques, but still significant for using transfer learning for the same reason as our study needed.

Hyperparameter Tuning

Despite the strong performance of Mask R-CNN in object detection and segmentation tasks, it still faces several limitations that affect its generalization and accuracy, especially in complex environments like forests (He et al., 2017). One of the challenges is its reliance on predefined anchor boxes, which may

not suit the irregular shapes and varying sizes of epiphytes and parasitic plants on host trees (Kong et al., 2019). Additionally, Mask R-CNN requires careful tuning of hyperparameters such as learning rate, batch size, and number of epochs to prevent overfitting or underfitting the model (Krizhevsky et al., 2012). Without proper optimization, the model may produce segmentation masks that are inaccurate or misaligned, especially when dealing with overlapping or partially occluded objects (Zhao et al., 2019). Therefore, while Mask R-CNN provides a powerful framework, it still demands specific optimization techniques to maximize its performance in environmental applications.

To address these issues, different optimization techniques for tuning Mask R-CNN's hyperparameters have been proposed to improve accuracy and reduce computational cost (Li et al., 2020). Other hyperparameter tuning methods used to find suitable values for hyperparameters, but they are often time-consuming and inefficient for large models like Mask R-CNN (Bergstra & Bengio, 2012). On the other hand, advanced optimization approaches in the study have shown better performance by intelligently exploring the search space and reducing training time (Snoek et al., 2012). These techniques are especially useful in ecological datasets where data imbalance and noise can affect the learning process (Shorten & Khoshgoftaar, 2019). Thus, selecting the right optimization strategy is crucial to making Mask R-CNN more adaptive and effective for detecting multiple species in natural habitats.

Moreover, Mask R-CNN struggles with detecting small objects, which is a common challenge when identifying small epiphytes or early-stage parasitic plants on tree trunks and branches (Zhao et al., 2019). This limitation affects the model's recall rate and precision, which are important metrics for evaluating

detection performance in real-world scenarios (Ren et al., 2015). Furthermore, its computational requirements are relatively high, making it less suitable for edge devices or mobile deployments in forest areas without high-end processing power (Chen et al., 2019). As a result, optimizing both the network architecture and its hyperparameters plays a key role in making the model more suitable for practical ecological monitoring (Kong et al., 2019). Hence, improving Mask R-CNN through optimization directly contributes to enhancing multi-object detection systems used for understanding plant symbiosis.

Hyperparameter tuning aims to find the optimal set of hyperparameters for a deep learning model to maximize its performance. The combination of dynamic search space definition and asynchronous pruning and sampler plug-ins in hyperparameter tuning technique enables to optimize learning rates and batch sizes and backbone selections and regularization parameters at large scales (Akiba et al., 2019). The multi-objective HPO literature shows researchers can optimize both mean average precision (mAP) and latency simultaneously for real-time detection applications (Karl et al., 2022). Furthermore, this study (Egorov & Reznik, 2021) used different numbers of epochs (5, 10, and 15) to determine whether the model's performance was improving or not, as there are cases where increasing the number of epochs does not lead to better results; therefore, it is important to evaluate it every training.

Recent object detection research demonstrates the effectiveness of systematic HPO through hyperparameter optimization. In other study in brain vessel analysis optimized base learning rate and maximum iterations and batch size to achieve their best results at 1e-2 learning rate with 10,000 iterations and

batch size 128 which produced AP@0.50 of 0.989 and AP@0.75 of 0.702 and AR of 0.802 (Kaya et al., 2024). The results validate previous research which demonstrates that thorough hyperparameter optimization leads to significant improvements in average precision (Raiaan et al., 2024). The optimization of Mask R-CNN parameters including learning rate and batch size and weight decay and ResNet-101 backbone settings resulted in 91% accuracy for complex overlapping objects in construction and demolition waste (CDW) segmentation compared to U-Net's lower performance (Shanthi & Rampradheep, 2025). The results match previous research which shows that Mask R-CNN and similar instance-level models require specific optimization approaches and backbone configurations to achieve their best performance (He et al., 2017).

The direct implementation of hyperparameter tuning in computer vision tasks proves its capability to discover optimal learning schedules and training budgets which result in better accuracy and improved generalization performance (Akiba et al., 2019a). The study used optimization technique to optimize VGG16 transfer learning parameters for seed disease prediction in agricultural imaging by adjusting learning rate and batch size and number of epochs. The implementation of hyperparameter optimization led to substantial performance enhancements in all evaluation metrics including accuracy and precision and recall and F1-score for image-based tasks with restricted prior knowledge (Bhadana & Kurunthachalam, 2025). The optimization of YOLO variants through hyperparameter tuning methods leads to enhanced detection accuracy and mAP performance according to recent research (Dahiya et al., 2023; Hoang et al., 2025).

The performance of Mask R-CNN depends on its detector and mask heads and ROI alignment according to its original formulation which means small hyperparameter adjustments affect multiple model components (He et al., 2017). The optimization of hyperparameters and backbone selection and non-maximum suppression methods in instance segmentation tasks leads to better results for road-scene analysis and stacked electronic component detection (Wan et al., 2023; Yang et al., 2020). The combination of multi-objective optimization allows researchers to achieve maximum accuracy (mAP) and operational efficiency which makes models ready for deployment and large-scale annotation work (Karl et al., 2022).

SYNTHESIS AND RELEVANCE OF THE STUDY

Monitoring forest ecosystems through the detection of symbiotic organisms such as epiphytes and parasitic plants is essential to assess ecological health and biodiversity. Traditional methods of monitoring these organisms are mostly manual, time-consuming, and prone to inaccuracies due to human limitations, especially in complex environments like forest canopies. With the increasing demand for real-time and accurate data, deep learning-based models have become significant tools for automating and enhancing the process of detection. Specifically, object detection algorithms like Mask R-CNN provide pixel-level accuracy and segmentation capabilities, making them suitable for ecological applications where species differentiation is crucial. This study aligns with the need to modernize ecological monitoring by integrating deep learning into environmental systems.

Additionally, the study emphasizes the importance of symbiotic relationships in forests, where the epiphyte plants benefit by gaining better access to sunlight, air, and moisture from their host tree, while the host tree is neither significantly helped nor harmed, while parasitic plants derive nutrients from the host, often causing it harm. Detecting these organisms accurately enables better forest management and contributes to conservation efforts. However, challenges in visibility, data imbalance, and misidentification in overlapping organisms pose a limitation to manual classification. As a solution, object detection using deep learning offers a way to efficiently process and analyze large image datasets with increased precision. Integrating Mask R-CNN, with its advanced segmentation and detection capabilities, is a logical step in addressing these challenges and filling the gap left by manual survey techniques.

Object detection has evolved rapidly with the development of one-stage and two-stage deep learning models. While one-stage detectors like YOLO are designed for speed, two-stage detectors such as Faster R-CNN and Mask R-CNN provide higher accuracy—an essential requirement for detecting small and occluded objects like epiphyte plants and parasitic plants. The reviewed studies show that although YOLO performs well in real-time scenarios, Mask R-CNN outperforms it in precision and segmentation, especially in complex scenes with multiple objects. Given that this study prioritizes accuracy and instance-level segmentation processing, Mask R-CNN was selected and further optimized for detecting multiple organisms on host trees.

Moreover, the study indicates that Mask R-CNN can perform well even with small datasets when combined with proper techniques like data

augmentation, hyperparameter tuning, and optimization algorithms. This is particularly relevant since datasets involving rare plant species in forest environments are often limited. The research also highlights that optimizing Mask R-CNN's architecture and training parameters can address its limitations in detecting small and overlapping objects in different characteristics of epiphytes and parasitic plants. Therefore, the study's focus on using an optimized version of Mask R-CNN is supported by recent developments and findings in the field.

Overall, by combining insights from ecology, computer vision, and deep learning, this study contributes to both environmental science and technological innovation. The application of a deep learning-based multi-object detection model for symbiotic organisms bridges a significant research gap between environmental monitoring. Through this approach, it not only enhances detection accuracy but also supports long-term ecosystem observation with minimal human intervention. The findings are expected to have practical implications in forest conservation, biodiversity research, and the development of intelligent monitoring systems.

RELATED LITERATURE MATRIX

Table 1 illustrates a comprehensive overview of studies from various academic literatures exploring different methodologies, algorithms, and findings for support in this study. Understanding how these plants interact with their host trees helps in monitoring forest health and protecting the environment. Due to human error activities and climate change affect forests, finding these symbiotic relationships early is important for managing and maintaining ecosystems.

Furthermore, the matrix also points out the gaps with traditional monitoring methods. Observing plants in forests, especially those high in trees, takes a lot of time and can lead to mistakes. These issues show the need for a more accurate and automated way of identifying species. Deep learning-based object detection model can help solve these problems.

The matrix also includes a detailed and comprehensive review of the existing literature deep learning models and object detection with Mask R-CNN algorithm that are highly relevant to the study. One-stage detectors give faster results but are less accurate, while two-stage detectors are slower but more precise. Since this study focuses on detecting small and overlapping objects, two-stage models like Mask R-CNN are more suitable. Using convolutional neural networks and image segmentation also allows the model to detect multiple objects in one image.

Several studies also show that models can still work well even with a small number of training images. Through methods like data augmentation, transfer learning, and tuning of parameters, the models can perform effectively. This is important for this study because collecting a large number of forest images is difficult and time-consuming. These findings support the use of an optimized model for model development.

Lastly, the literature matrix also shows improvements in model architecture and optimization strategies that directly enhance performance. Better backbone networks and tuning methods help detect small or partly occluded objects. These ideas support the goal of this study to create a reliable model for detecting epiphyte plants and parasitic plants using Mask R- CNN.

Table 1. Matrix of Reviewed Related Literature

AUTHOR/S	TITLE	YEAR	ALGORITHM/ TECHNIQUE	FINDINGS
He et al.	Mask R-CNN	2017	Mask R-CNN	Achieved 38.2% mAP on COCO dataset using ResNet-101; showed accurate instance segmentation with pixel-level masks, enabling object detection and segmentation.
Afzaal et al.	Deep Learning-Based Instance Segmentation of Strawberry Diseases	2021	Mask R-CNN with ResNet backbone	Achieved 82.43% mAP under complex backgrounds for detecting strawberry diseases; model successfully separated overlapping objects.
Musyarofah et al.	Roof Detection on Aerial Images using Mask R-CNN	2020	Mask R-CNN	Achieved mAP of 90.14% and precision of 88.71% using RGB features for aerial imagery detection.
Hou & Li	Building Feature Extraction using Mask R-CNN	2024	Mask R-CNN	Achieved 95% classification accuracy, outperforming CNN, R-CNN, and GAN in aerial building extraction tasks.
Siricharoen et al.	Robust Object Detection using Small Datasets	2023	Mask R-CNN	Achieved strong accuracy even with a small dataset; introduced optimal thresholding for improved robustness.
Wu et al.	RGC Mask R-CNN: Improved Object Detection Method	2020	RGC Mask R-CNN	Achieved superior accuracy using ResNeXt-101, group normalization, and 3-stage cascade IoU thresholds; tested on COCO and VOC07.
da Cruz et al.	Soybean Foliar Disease Detection using Mask R-CNN	2025	Mask R-CNN with ResNet-101	Used only 525 images, augmented to 2,408; still achieved high detection accuracy for foliar diseases.
Zikra et al.	Instance Segmentation with Limited Data using Mask R-CNN	2023	Mask R-CNN	Achieved high mAP with just 651 images and 8,922 annotated objects, showing effectiveness even with a limited dataset.
Islam et al.	CB-Net for Ecological Detection	2024	Mask R-CNN + CB-Net + Transfer Learning	Used 1,000 images with ResNet-101 and transfer learning to reduce processing time while maintaining accuracy.
Zhou et al.	Slot-Based Image Augmentation for Detection	2020	Mask R-CNN + Augmentation	Improved model performance with novel augmentation method, addressing limited data challenges in object detection.
Chowdhury et al.	Transfer Learning for Small Dataset Object Detection	2024	Mask R-CNN + Transfer Learning	Proved efficiency in reducing training time and enhancing accuracy using smaller datasets.
Kong et al.	Limitations in Anchor-based Detection	2019	Mask R-CNN	Identified performance drops when detecting irregular and overlapping shapes like epiphytes, suggesting need for anchor-free methods.
Chen et al.	Mask R-CNN for Agricultural Pest Monitoring	2021	Mask R-CNN	Applied to pest detection; achieved over 80% precision in identifying pests under varying lighting and crop stages.

CONCEPTUAL FRAMEWORK

This study begins with the acquisition of image data focusing on host trees and the plant organisms involved in plant symbiosis, specifically, epiphytes and parasitic plants. Since publicly available datasets are limited in quantity and quality, the majority of the dataset is built through manual photography in natural outdoor settings. These images reflect real-world conditions such as natural lighting, variable backgrounds, and realistic plant positioning.

Once the images are collected, the dataset undergoes data processing, which includes two key steps: annotation and augmentation. In the annotation stage, the epiphytes, parasitic plants, and host trees are labeled manually to prepare the data for training. In parallel, data augmentation techniques such as flipping, rotating, cropping, and brightness adjustments are applied to artificially expand the dataset and increase variability. This helps reduce overfitting and improves the model's ability to detect objects in diverse environmental conditions.

The annotated and augmented dataset is then used to implement the Mask R-CNN model, a deep learning algorithm that performs object detection, instance segmentation, and classification. The model is trained to identify the three target classes in a single image. After the initial training, the model is evaluated using standard performance metrics such as precision, recall, F1-score, mean Average Precision (mAP), Intersection over Union (IoU), and confusion matrix.

Once initial results are gathered, the gaps in the model's performance are then reviewed. Based on this review, the model may be optimized further

by exploring adjustments in parameters, refining preprocessing methods, improving data augmentation strategies, or even optimizing its backbone network to enhance detection accuracy and reliability. This step aims to improve the model's accuracy and robustness when dealing with real-world image complexities.

In summary, this conceptual framework highlights a focused pipeline, from real-world data collection and annotation to deep learning implementation and performance-based optimization. The goal is to build a reliable and accurate model for detecting epiphytes and parasitic plants on host trees, contributing to advancements in ecological monitoring and applied computer science.

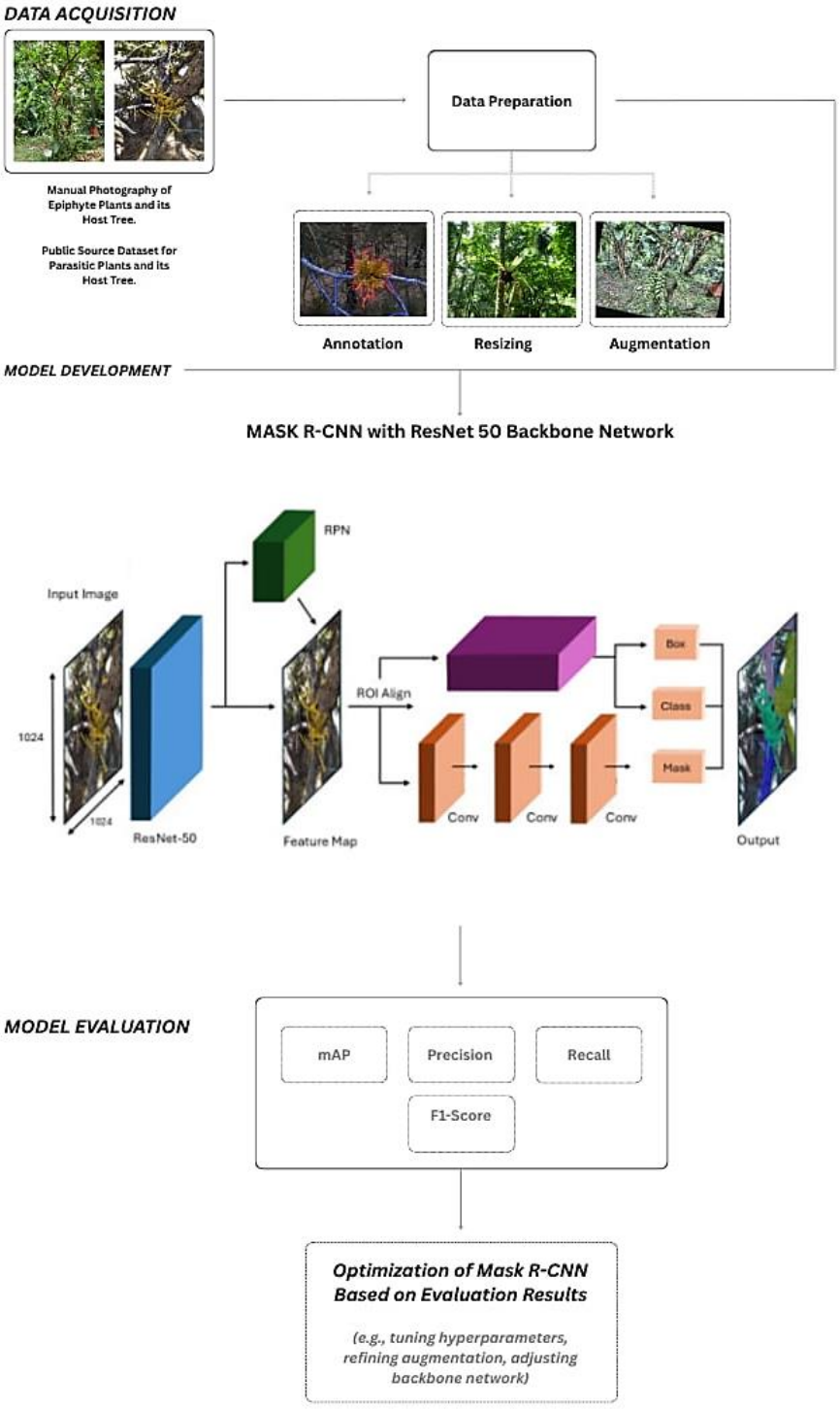


Figure 1. Conceptual Framework for Deep Learning-Based Multi-Object Detection of Epiphytes and Parasitic Plants on Host Trees Using Optimized Mask R-CNN for Symbiotic Relationship

CHAPTER III

METHODOLOGY

RESEARCH DESIGN

The research design employed theoretical-experimental methods to create and test a deep learning model that identifies epiphyte plants, parasitic plants, and host trees. The experimental section of the study employed the Mask R-CNN architecture to train and evaluate a baseline model using standard hyperparameters parameters. The researchers developed an optimized Mask R-CNN model by adjusting the data preparation process, hyperparameter tuning, and model configurations, which involves modifying the backbone network from ResNet50 to ResNet101, training hyperparameters including the learning rate, weight decay, batch size, and number of epochs.

The researchers applied quantitative performance metrics to evaluate detection and segmentation quality for both baseline and optimized model assessment. The research design enables both descriptive-developmental model development and documentation and analytical-comparative performance evaluation through model performance assessment. The research design achieved its goal by demonstrating how optimization enhanced Mask R-CNN performance in detecting and segmenting symbiotic plant relationships.

STUDY AREA

This study was conducted at the Makiling Botanic Gardens (MBG), located within the University of the Philippines Los Baños (UPLB) campus in Laguna. This area was specifically chosen because of its rich biodiversity and well-preserved natural environment, which provides an ideal setting for

observing epiphytes, parasitic plants, and their host trees in actual symbiotic interactions.

Makiling Botanic Gardens is part of Mount Makiling Forest Reserve, a recognized ASEAN Heritage Park, making it a highly valuable site for ecological research. The reserve supports a wide range of native and endemic flora, including various species of epiphytes like orchids and ferns, and parasitic plants such as mistletoes. The diversity and abundance of plant species in MBG offer a realistic and extensive dataset for training and testing the deep learning model.

Furthermore, the location is accessible to the researchers and permits were obtained to carry out the image collection in a controlled and sustainable manner. Collecting real-world images in a scientifically significant and ecologically diverse location, the study ensures that the dataset is not only rich in content but also relevant to actual forest conditions, thereby increasing the potential applicability of the deep learning model in future conservation and ecological monitoring efforts.

MATERIALS

This study was conducted that requires the use of both hardware and software resources that played a crucial role in developing and implementing the model. The hardware provided the necessary computational power to handle intensive training and testing processes, while the software offered the tools and frameworks essential for developing, optimizing, and evaluating the model's performance. Together, these ensured that the study could be carried out effectively and with reliable results.

HARDWARE

This study used a laptop and a DSLR camera as the main hardware tools. The laptop was used for writing code, processing data, and testing the deep learning model. It had a 13th Gen Intel(R) Core(TM) i5-1335U processor running at 1.30 GHz, with 16.0 GB of RAM (15.7 GB usable), and a 64-bit operating system with an x64-based processor. These specifications were sufficient to handle the tasks involved in developing the project, such as preparing the dataset, annotating images, and writing scripts. However, because training deep learning models required more processing power, especially with large image data, cloud-based platforms like Google Colab were used for those heavier tasks.

For collecting images, a Canon EOS 750D DSLR camera was used. This camera had a 24.2-megapixel APS-C sensor, a 19-point cross-type autofocus system, and continuous shooting of up to five frames per second. It was chosen because it could capture clear and detailed images, which were important for detecting small plant features in outdoor conditions. Its fast autofocus and shooting speed were also helpful during manual image collection, especially when capturing multiple angles or when lighting conditions changed. Overall, the combination of the laptop and DSLR camera provided the necessary hardware to support both data collection and system development.

SOFTWARE

Software materials such as cloud-based platforms, data science platform, programming languages, source-code editors, frameworks, and supporting libraries that were used in this study played a significant role in the development and testing of the deep learning models.

Cloud-Based Platform

Google Colab is a cloud-based platform that allowed users to run Python code using free access to GPUs (graphics processing units). It also worked well with Google Drive, which made it easy to access and save datasets and files. Another reason why Google Colab was chosen was because it was beginner-friendly and supported many of the tools and libraries needed for deep learning.

This platform was used during the training and testing of the deep learning model. It was very helpful because training a deep learning model took a lot of time and computing power, which personal computers might not have been able to handle. By using Google Colab, the researchers were able to train the model faster and more efficiently. It was also important since there was limited hardware. Instead of relying only on laptops or desktop computers, Google Colab provided a more flexible and powerful environment to experiment, test, and improve the model without worrying about system crashes or slow performance.

Platform for Data Science

Roboflow was a platform that was used to build and deploy computer vision models. It offered tools for data management, annotation, model training, and deployment, making it easier to develop custom computer vision solutions. Uploading, managing, annotating, and analyzing images and videos, training custom models using state-of-the-art technologies, and deploying those models in the cloud or on local hardware were the steps that could be done in Roboflow.

Programming Language

Python programming language was widely known for its simplicity and versatility, especially in machine learning and deep learning applications.

Python supported a wide range of libraries that were specifically built for tasks like data processing, model training, evaluation, and visualization. Its syntax was beginner-friendly, which made it easier for researchers to experiment and troubleshoot during model implementation. Furthermore, Python had a large community and documentation support, which was a great help when dealing with bugs or unfamiliar functions.

Source Code Editor

Visual Studio Code (VS Code) is a lightweight and powerful source-code editor developed by Microsoft that supports various programming languages, including Python, which is widely used in deep learning and artificial intelligence. It serves as the primary coding environment for developing, testing, and debugging deep learning models, providing features such as syntax highlighting, intelligent code completion, and integrated version control. Through its extensive extensions such as Python and Git.

FRAMEWORKS AND LIBRARIES

PyTorch

PyTorch serves as the primary deep learning framework used in the study. It provides a flexible tensor computation environment and supports automatic differentiation, which facilitates the training and optimization of convolutional neural networks. Its modular structure allows for the efficient implementation and customization of deep learning architectures.

Detectron2

Developed by Facebook AI Research (FAIR), Detectron2 is a high-performance computer vision framework built on top of PyTorch. It is specifically designed for object detection, instance segmentation, keypoint detection, and

panoptic segmentation tasks. Detectron2 simplifies model training and evaluation by providing pre-configured architectures, such as Mask R-CNN, and built-in utilities for dataset management, augmentation, and performance evaluation.

OpenCV (cv2)

OpenCV, or Open Source Computer Vision Library, is used for image manipulation and preprocessing. It enables operations such as image loading, resizing, color conversion, and visualization. Within this study, OpenCV assists in handling image inputs, generating segmentation masks, and visualizing detection outputs.

Pycocotools

Pycocotools is a Python API used to work with the COCO (Common Objects in Context) dataset format. It facilitates reading, writing, and evaluating COCO-style annotations, including bounding boxes and segmentation masks. In this research, it is primarily employed to compute COCO-based evaluation metrics such as mean Average Precision (mAP).

NumPy

NumPy provides efficient numerical operations and array manipulation capabilities. It is used to perform matrix computations, process image pixel values, and handle intermediate data representations between model predictions and evaluation functions.

Pandas

Pandas is utilized for data handling and analysis. It allows the structured storage of results, metrics, and annotations in tabular form, which facilitates

easier analysis and export of performance data such as accuracy, precision, recall, and F1-scores.

Matplotlib and Seaborn

Matplotlib and Seaborn are visualization libraries used to present model performance graphically. Matplotlib is responsible for generating line plots of training losses and performance metrics, while Seaborn enhances these visualizations with styled heatmaps and statistical plots, such as confusion matrices.

Scikit-learn

Scikit-learn provides additional statistical evaluation tools. It is employed to compute precision, recall, F1-score, and confusion matrices from model predictions. These metrics are essential in quantitatively assessing the performance and reliability of the object detection and segmentation model.

COCOEvaluator (from Detectron2)

The COCOEvaluator module is a built-in Detectron2 component used to assess model performance using COCO-style metrics. It automates the computation of key indicators such as mAP at different Intersection over Union (IoU) thresholds, providing standardized and comparable evaluation results.

OS and JSON Libraries

The OS library is used to manage file paths, directories, and system-level operations, while the JSON library handles reading and writing of structured data files, including training logs and configuration files. Together, they streamline dataset management and result organization.

DATA COLLECTION METHOD

In this study, the image dataset was compiled through a manual data collection and publicly available dataset and. Since our target parasitic plant was not discovered in our study area, we took it from a public dataset; however, a manual approach to data collection was also performed because there was no publicly available dataset focused on specific epiphytes.

We collected epiphytes manually because the target data are present in the study area. Manual images were taken using a DSLR camera, which allowed us to capture high-quality images of various plants in their natural environment. A DSLR camera was used because it can focus well on small plant features and handle outdoor lighting conditions better than a phone camera. This helped us get clear and detailed images, which is essential for training deep learning models like Mask R-CNN.

We took multiple images from different angles and distances to ensure that objects were well represented in the dataset. The manual process took time and effort, but it was necessary to build a unique dataset that matched our study. By collecting our own data, we ensured that the images were accurate, up-to-date, and truly represented the symbiotic relationship that is related to the study.

MODEL DEVELOPMENT

This section outlines the development process of the proposed study, which focuses on implementing a deep learning-based object detection model, Mask R-CNN, to detect multiple plant organisms involved in plant symbiosis, specifically epiphyte plants, parasitic plants, and their host trees. The model was trained and evaluated using a custom dataset consisting of manually

captured images obtained from natural outdoor settings as well as online resources. After its initial implementation, the model's performance was reviewed and optimized based on the results of evaluation metrics.

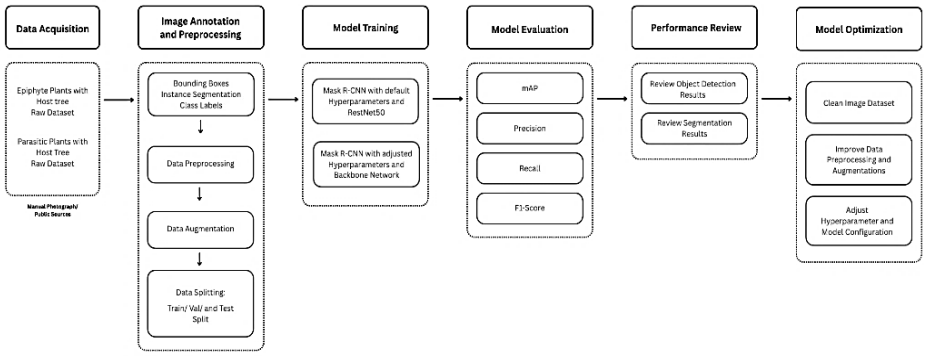


Figure 2. Model Development Diagram

DATA ACQUISITION

The dataset used in this study consists of three class categories, which are epiphyte plants, parasitic plants, and host trees. Initially, 212 images of epiphytes with host tree were collected through manual photography in Makiling Botanic Garden, and 344 images of parasitic plants with host tree were collected in a publicly available dataset. These methods were performed due to the limited availability of specific parasitic plants, so public datasets are relevant to the study. Images reflect real-world conditions such as natural lighting, varied plant structures, and organic growth patterns. Since the dataset is collected both manually and from public sources, there is a possibility of class imbalance, and to address this, additional augmentation was applied selectively to underrepresented classes during preprocessing.

Table 2. Raw Image Dataset Collected

Class Name	Images Collected
Epiphytes with Host Tree	212
Parasitic Plants with Host Tree	344
Total	556

Epiphyte Plants

Epiphytes were plants that grew on other plants, typically trees, without drawing nutrients from them. They relied on the surrounding environment, such as air, rain, and debris, for survival. In this study, epiphytes were included because they formed a common symbiotic relationship with trees and were often indicators of environmental health. The collected images featured two species of epiphytes in their natural habitat, which is Bird's Nest Fern and Sanggumay (Dendrobium anosmum), that can be seen both in controlled and open environments growing on trees.

Parasitic Plants

Parasitic plants, on the other hand, formed a contrasting relationship with their hosts. These plants attached themselves to trees and extracted nutrients, potentially causing harm to the host trees. Their inclusion in the dataset was essential for training the models to distinguish between neutral and harmful plant interactions. The parasitic plant species is commonly known as Dwarf Mistletoes.

Host Tree

The host trees served as the primary surface or base where both epiphytes and parasitic plants were typically found. This was important for helping the model recognize and learn the natural texture, patterns, and features of tree surfaces. Including these neutral or unoccupied images improved the model's ability to differentiate between the presence and absence of symbiotic organisms, thus reducing false detections and enhancing overall classification accuracy.

Image Annotation, Augmentation, and Preprocessing

After collecting and organizing the dataset, the next phase would be preparing the images for training through annotation, augmentation, and preprocessing. These steps are crucial to ensure that the deep learning model can correctly learn and identify the relevant objects in each image and make it robust.

Image Annotation

To prepare the dataset for Mask R-CNN training, a detailed image annotation process was conducted, focusing on accurately labeling specific regions of interest, which included epiphytes, parasitic plants, and host trees. This step was critical for helping the model distinguish between the different object classes based on their visual characteristics.

Annotation was performed using Roboflow, an accessible and widely used tool that supports instance segmentation. Since Mask R-CNN works with pixel-level segmentation, each object in an image was annotated with a polygon mask outlining its exact shape and size. These instance masks allowed the model to learn not only the presence and location of an object but also its boundaries and structure.

During this process, special attention was given to the detailed features of each class. Consistency and accuracy in labeling were maintained to reduce noise and improve the model's ability to generalize to new images captured in natural outdoor conditions.

Data Preprocessing

To enhance the model's performance and generalizability, data preprocessing techniques are applied to the annotated dataset. All images are

resized to 1024x1024 resolution that matches the input requirement of the Mask R-CNN architecture. This step ensures consistency in training and avoids issues related to varying image dimensions. In addition to resizing, data augmentation techniques are used to artificially increase the diversity of the dataset and address the limitation of having a small number of original images and instances. These augmentation methods include horizontal flipping, rotation, brightness, saturation, exposure, blue and noise adjustments, all of which help the model become more robust when exposed to different object orientations, lighting conditions, and visual variations during inference.

Data Splitting

After annotation and preprocessing, the dataset is divided into three subsets: training, validation, and testing. The training set is used to allow the model to learn from the data, while the validation set is used to tune the model's performance and monitor issues such as overfitting during training. The test set consists of unseen images used for the final evaluation to ensure that the model can generalize well to new data. The dataset was split approximately into 70% training, 15% validation, and 15% testing to ensure sufficient learning while preserving evaluation integrity. This structured splitting process supports reliable performance measurement and ensures that the model's evaluation is based on independent input not previously encountered during training.

Table 3. Total Number of Annotated, Augmented, and Preprocessed Images

Split	Number of Images
Training Split	1467
Validation Split	97
Test Split	97
Total	1661

Table 4. Total Instances of Classes

Class Name	Instances
Epiphyte Plants	2539
Parasitic Plants	2387
Host Tree	2970
Total	7896

MODEL TRAINING

In this study, the selected deep learning architecture was Mask R-CNN, initially implemented with a ResNet-50 backbone and default hyperparameters, which served as a baseline for learning object features. If the model's performance during evaluation indicated challenges in object detection and segmentation, a deeper backbone architecture, such as ResNet-101, together with various adjusted hyperparameters, was introduced in the optimization phase to enhance feature representation and overall detection performance. The model was implemented using the Detectron2 framework in Python, which supports instance segmentation, flexible backbone integration, and model hyperparameter configuration.

The training process was carried out using the manually prepared dataset, which had been split into training, validation, and testing subsets. The training set allowed the model to learn from examples, while the validation set helped monitor the model's performance and detect issues such as overfitting. The test set remained untouched during training and was later used for final performance evaluation.

Since Mask R-CNN is capable of instance segmentation, it provided precise object boundaries, an essential feature for detecting plant structures that are often irregular, overlapping, or partially obscured. After initial training, the model's output was reviewed to observe potential detection challenges or

segmentation inconsistencies. These findings guided further improvements in the model, which included tuning hyperparameters such as learning rate or batch size, optimizing data preprocessing strategies, and adjusting the model configuration.

MODEL EVALUATION

Once the model is trained, its performance is evaluated using standard object detection and segmentation metrics to understand how well it can detect, segment, and classify each object. These include mean Average Precision (mAP), precision, recall, F1-score, Intersection over Union (IoU), and the confusion matrix. These metrics help measure the model's ability to correctly detect objects, such as true positives, false positives or false negatives.

Evaluation was conducted on a separate set of images that were not included in the training process to test how well the model generalized to new and unseen data. If the model failed to meet acceptable benchmarks—for example, if the mAP was below 0.5 or there was frequent confusion between classes—optimization techniques were initiated. This step helped confirm the model's effectiveness and reliability in identifying plant symbionts and host trees in real-world conditions.

Mean Average Precision (mAP)

Mean Average Precision (mAP) gave an overall score that summarized the model's ability to both correctly detect and accurately locate objects. It combined precision and recall results across different classes and different IoU thresholds. In this study, mAP was calculated at both mAP@0.5 and mAP@0.5:0.95. This helped fairly judge the model's performance under both loose and strict conditions, which was important because detecting epiphytes

and parasitic plants on their host trees was challenging due to overlapping leaves, branches, and other visual noise in the images.

Precision

Precision showed how many of the detected objects were actually correct. It indicated how well the model avoided false detections. A high precision score meant that when the model detected an epiphyte or parasitic plant, it was likely correct. This was important because incorrect detections could lead to errors in analyzing the symbiotic relationship. Precision was calculated for each class using intersection-over-union (IoU) to ensure that the predicted mask properly matched the actual object.

Recall

Recall measured how many of the actual objects present in the image were correctly detected by the model. A high recall meant that the model could find most of the epiphyte plants and parasitic plants, even if they were small, hidden, or partly covered by other parts of the tree. This helped ensure that the model did not miss any important objects, which was essential for accurately detecting and identifying epiphytes and parasitic plants on canopy trees.

F1-Score

The F1-score provided a balanced measurement between precision and recall. It was useful to ensure that the model not only avoided false detections but also did not miss actual objects. A high F1-score indicated that the model performed well in both precision and recall simultaneously.

Intersection over Union (IoU)

IoU measured how close the predicted mask and bounding box were to the ground truth. It was calculated by dividing the area of overlap between the

predicted region and the actual region by the total area covered by both. A higher IoU indicated that the model accurately located the object. In this study, IoU was used to evaluate whether the detected epiphytes and parasitic plants were not only correctly identified but also well-aligned with their actual shape and position on the host tree surface.

Confusion Matrix

The confusion matrix was used to count how many times the model correctly or incorrectly classified each object class. It included values such as true positives, false positives, and false negatives. This matrix helped analyze where the model made mistakes, such as confusing parasitic plants with epiphyte plants or missing detections.

MODEL OPTIMIZATION

After the initial training and testing phases, the model's performance was reviewed to identify potential limitations, such as detection inaccuracies, segmentation errors, or confusion between classes. Based on these observations, the model underwent further optimization to enhance its overall performance.

One of the first steps in this process was evaluating the backbone network. Initially, the model was implemented using ResNet-50, but if performance issues occurred, such as poor feature extraction or missed detections, a shift to a deeper backbone like ResNet-101 was considered to improve accuracy and representation learning.

Other optimization efforts may involve refining data preprocessing steps, improving data augmentation strategies, or revisiting model configuration settings. These iterative adjustments are intended to improve the model's robustness and accuracy, especially when dealing with the natural variability and visual complexity found in outdoor forest environments.

EXPERIMENTAL SETUP

The experimental setup for this study was organized to ensure reliable and measurable training behavior of the Mask R-CNN model for detecting and segmenting epiphytes, parasitic plants, and host trees. The setup includes dataset preparation, preprocessing, and augmentation procedures, and a structured training process, which systematically evaluates different learning rates, epoch settings, and backbone architectures. The overall workflow of the experiment is illustrated in the diagram below.

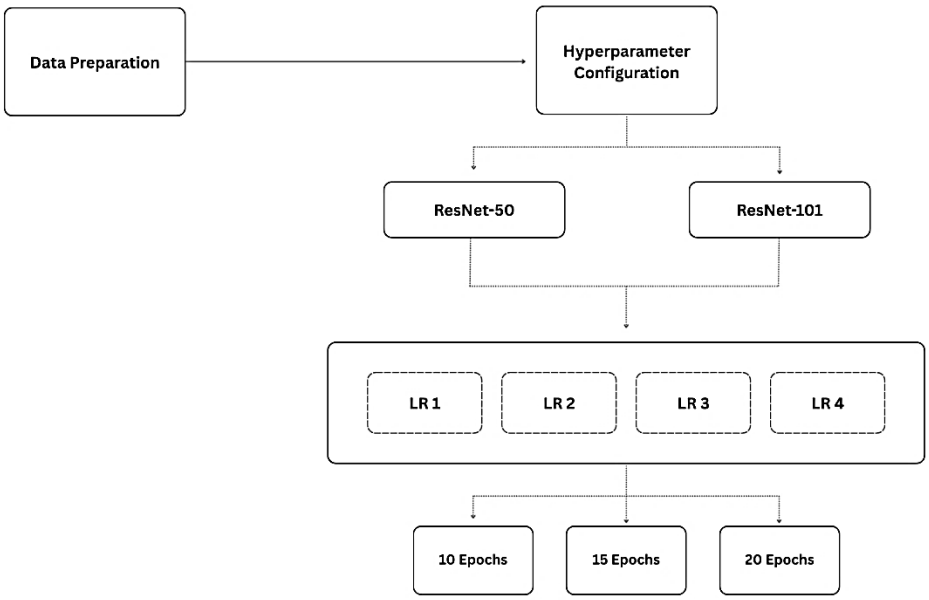


Figure 3. Experimental Setup Diagram

The dataset was divided into a 70/15/15 ratio to provide balanced learning and unbiased evaluation. Preprocessing and augmentation were applied to enhance image clarity and increase sample diversity, allowing the model to adapt to natural variations in plant appearances and background environments. After dataset preparation, the Mask R-CNN model was trained following a strictly structured process designed to identify the best-performing configuration for this study. The model was trained using Detectron2 with two data loader workers and a batch size of two images per iteration. A weight decay value of 0.0001 was set to prevent overfitting, while 200 warm-up iterations ensured gradual and stable learning. The learning rate schedule was configured at 4,453 and 6,012 steps to maintain controlled optimization. The model was defined with three classes corresponding to the target plant symbiotic components. Evaluation occurred at the end of every epoch to consistently monitor loss behavior and performance improvement.

The structured training process was carried out in three main stages. First, four different learning rates (0.01, 0.005, 0.0025, and 0.00125) were independently tested. Second, each learning rate was trained under three different epoch durations: 10, 15, and 20 epochs. Third, the entire configuration was evaluated across two backbone architectures, starting with ResNet-50 and followed by ResNet-101 to determine whether deeper feature extraction improves segmentation outcomes. This systematic approach ensured that every possible training variation was examined, allowing the study to establish the most effective combination of hyperparameters and network depth for segmenting epiphytes and parasitic plants on host trees.

ETHICAL CONSIDERATION

This study was conducted with careful attention to ethical standards, particularly in data collection, image usage, and model development. The dataset used in the experiment comprised images of epiphytes, parasitic plants, and host trees, sourced from publicly available databases as well as manually collected field images. All image data were either in the public domain or used under fair academic and research purposes, ensuring that no personal or sensitive information was involved. Additionally, all model training and experimentation were performed in a responsible manner. The study did not involve any human or animal subjects, nor did it pose risks to privacy, safety, or ecological systems. The research strictly adhered to ethical guidelines emphasizing transparency, replicability, and integrity in experimental design. Furthermore, care was taken to ensure that no misinformation or ecological misclassification was introduced through the model outputs. The intention of this research was to support sustainable agricultural practices and scientific understanding, and not to misinterpret or misapply the detection results in ways that could impact environmental or conservation decisions.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the objectives, results, and discussion of the study, supported by visual outputs and performance evaluations of the developed model. The study aims to implement a Mask R-CNN model capable of detecting and segmenting epiphyte plants, parasitic plants, and host trees from still images. The results from each experiment are discussed in detail to show the model's performance, improvements after optimization, and its effectiveness in identifying the target objects.

The Implementation of Mask R-CNN for Multiple Object Detection and Segmentation of Epiphyte Plants, Parasitic Plants, and Host Trees

This section presents the initial implementation of the Mask R-CNN model using the original dataset to perform multiple object detection and segmentation of epiphyte plants, parasitic plants, and their host trees. The purpose of this phase is to establish the model's baseline performance before any optimization strategies are applied. It describes the training setup used, including the model configuration and hyperparameters, followed by the results obtained from the initial experiments. These results serve as the foundation for identifying the challenges and limitations of the model and the areas that require optimization in the succeeding section.

The Initial Training Setup of Mask R-CNN

This subsection explains the training setup used for the initial implementation of the Mask R-CNN model. It includes the dataset configuration, preprocessing procedures, and the hyperparameters applied during training.

The goal of this setup is to ensure that the model is properly trained under consistent and controlled conditions, allowing for a fair evaluation of its baseline performance before optimization.

The initial raw dataset consisted of 453 images and was split into training, validation, and testing sets using a 70/15/15 ratio. Preprocessing and data augmentation were applied only to the training set, increasing the number of training images to 1,014. However, the validation and test splits were not affected by the preprocessing and augmentations.

Table 5. Total Number of Annotated, Augmented, and Preprocessed Initial Image Datasets

Split	**Number of Images**
Training	1467
Validation	97
Test	97
Total	1661

Table 6. Total Number of Instances per Class of Initial Datasets

Class Name	**Instances**
Epiphyte Plants	2539
Parasitic Plants	2387
Host Tree	2970
Total	7896

Table 7. Details of Augmentations Used in Initial Model Training

Augmentations	Specifications
Flip	Horizontal
Rotation	Between -5° and +5°
Brightness	Between -10% and +10%
Exposure	Between -5% and +5%

Table 8. Details of Hyperparameters Used in Initial Model Training

Hyperparameter	Specifications
Learning rate	(0.02, 0.01, 0.005, 0.0025, 0.00125)
Weight Decay	0.0001
Batch Size	2
Epochs	10, 15, 20
Warmup Iterations	200
Number of Workers	2

Result of the Initial Implementation of Mask R-CNN

The initial implementation of the Mask R-CNN for multiple object detection and segmentation of epiphyte plants, parasitic plants, and host trees has achieved quite low performance in terms of the object detection and segmentation metrics. These results suggest an opportunity for improvement and optimizations that we were preparing for. The results from this are also significant as they would be the baseline for this study.

Table 9 presents all the results of mAP50 from various hyperparameters. First, we trained the model with its default hyperparameters, such as a learning rate of 0.02 and the backbone architecture ResNet-50, and the rest of the configurations. However, during the model training with the default 0.02 learning rate, it crashes out. We then proceed with training with four more learning rates to see which learning rate would get the best results from the combinations and the deeper backbone network ResNet-101. One of the goals of proposing different learning rates and model configurations is to spot which area of the hyperparameters is best and well-suited to this dataset.

After the initial implementation of the Mask R-CNN model, the results showed that modifying the hyperparameters and configurations gives the model better result metrics. Based on the table results, the default backbone architecture ResNet-50, achieved its best results in learning 0.01 and was trained on 10 epochs. Furthermore, by modifying its backbone architecture to ResNet-101, which is deeper than ResNet-50, it also showed an improvement for bounding box mAP50. Therefore, the initial implementation achieved its best results by modifying the backbone network from ResNet-50 to ResNet-101 with a learning rate of 0.01 in 10 10-epoch setup. This study proves that adjusting or modifying the hyperparameters improves its overall performance metrics.

However, based on the results table, when adjusting its epochs for each setup, it showed a decrease over time. So, in this initial implementation, 10 epochs is a well-suited duration for training the model, with a stable decrease in training loss and an increase in mAP every epoch. While adjusting the epochs, each setup showed a decrease, which also proves that the model overfits for longer training. Therefore, for the next phase of this study, it is

suggested that optimizing and adjusting the data preparation and some model configurations would be better for more experimentation for further improvements of the model.

Table 9. Results from the Initial Implementation in Mask R-CNN

Learning Rate	Epochs	RestNet-50		RestNet-101	
		Bbox mAP50	Segm mAP50	Bbox mAP50	Segm mAP50
0.01	10	73.379	67.307	73.088	67.694
	15	71.987	67.790	69.733	65.789
	20	68.243	64.033	69.534	65.970
0.005	10	71.893	65.973	72.063	66.389
	15	69.916	64.591	70.465	66.671
	20	70.554	66.311	69.548	63.749
0.0025	10	71.000	65.324	69.200	63.607
	15	70.163	64.406	69.948	64.713
	20	70.269	65.326	69.965	64.646
0.00125	10	70.755	64.310	71.225	64.320
	15	70.742	64.910	68.139	62.880
	20	69.384	63.102	66.407	61.657

Table 10 shows the results of the best performing initial implementation of the Mask R-CNN model. At the bounding box level, the model obtained an mAP score of 50.019 and an mAP50 score of 73.088. For segmentation, the model achieved an mAP of 34.714 and an mAP50 of 67.694. The precision, recall, and F1-score metrics were evaluated at the optimal confidence threshold of 0.8 and at IoU50. The bounding box results show a precision of 0.6839, recall

of 0.5934, and F1-score of 0.6346. Meanwhile, the segmentation level achieved a precision of 0.6185, recall of 0.5351, and F1-score of 0.5730. These values represent the performance of the model during its initial implementation stage.

Table 10. Results of the Best Performing Initial Implementation of Mask R-CNN

Metrics	Bounding Box Level	Segmentation Level
mAP	50.019	34.714
mAP50	73.088	67.694
Precision	0.6839	0.6185
Recall	0.5934	0.5351
F1-Score	0.6346	0.5730

Figure 4 presents the continuous decrease of training loss over the 10-epoch training, which suggests that the model is learning and not overfitting over the entire training process. Even with a slight increase in some of the iterations, which is an indicator of overfitting, it still found its way to learn and decrease the loss until the last epoch.

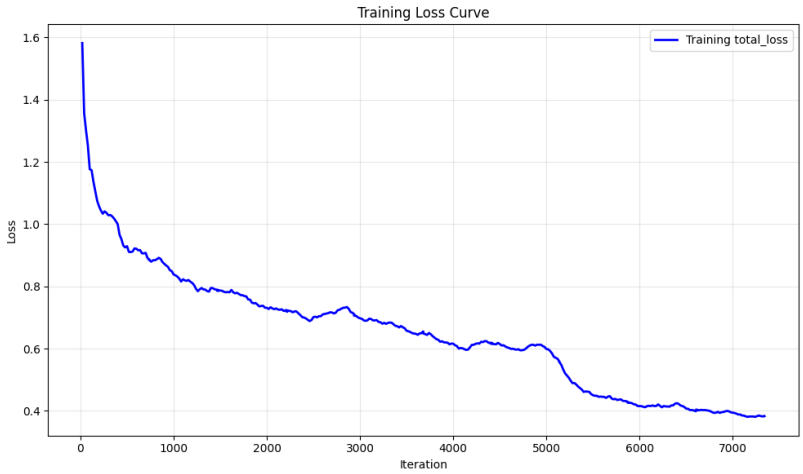


Figure 4. Training Loss of Best Result in Initial Implementation

Figure 5 presents the plot of validation mAP over the entire training process. Based on the plot result, it smoothly increases over each epoch, which is an indicator that the model is learning.

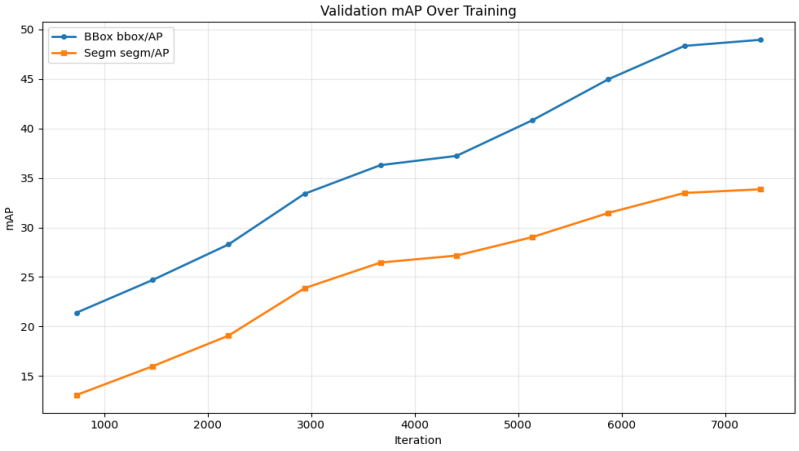


Figure 5. Validation mAP plot of Best Result in Initial Implementation

Figure 6 presents the confusion matrix for the bounding box level over the entire classes. The missed column represents the missed detection with each of the classes, and based on the table, it shows a minimal missed for each class. The missed row represents all the total confusion between each class, and based on the table, it just shows minimal confusion between each class.

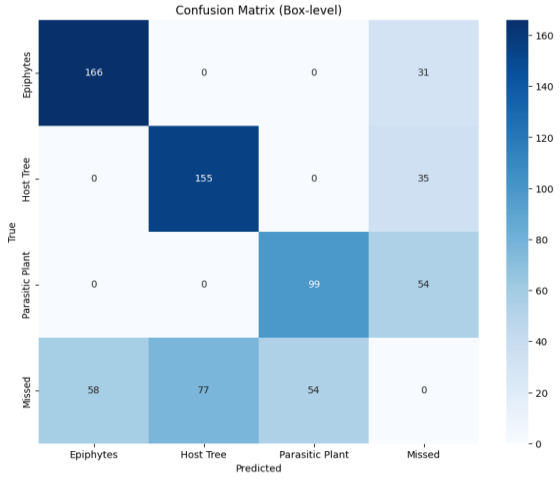


Figure 6. Confusion Matrix for Bounding Box Level of Best Result in Initial Implementation of Mask R-CNN

Figure 7 presents the confusion matrix for the segmentation level over the entire classes. The missed column represents the missed segmentation with each of the classes, and the missed row is for confusion between each class. The table results show quite misses and confusion compared to the bounding box level, as the segmentation level is way more difficult task.

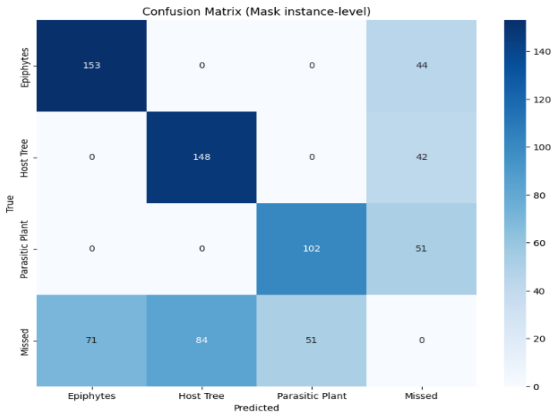


Figure 7. Confusion Matrix for Bounding Box Level of Best Result in Initial Implementation of Mask R-CNN

Performance Overview and Identification of Challenges

In this part, it overviews the model's performance after the initial implementation. Further observations are being carried out carefully to identify the challenges and weaknesses of the model in terms of detection and segmentation of the objects. Identified challenges would be significant for guiding the next phase of optimization of the model.

Figure 8 presents the performance of the initial implementation of Mask R-CNN for detecting and segmenting the epiphyte plants and the host tree. Based on the result, it correctly detects and segments the epiphyte plants and their host tree.



Figure 8. Overview of Object Detection and Segmentation of the Initial Output of Epiphyte Plants on Host Trees

Figure 9 presents the performance of the initial implementation of Mask R-CNN for detecting and segmenting the parasitic plants and the host tree. Based on the result, it correctly detects and segments the parasitic plants and their host tree, even with some partially occluded objects.

After a careful and strict review of the model's performance under the test set dataset, some of the potential challenges are identified. For epiphyte plants on host trees, the model missed some of the small and partially occluded plants in detection and segmentation. Some of the epiphyte plants also have a poor or are not fully covered by the mask because of the too complex angle of the plant. While in parasitic plants on host trees, it also missed some of the parasitic plants and their host trees. The detection and segmentation position is also too complicated, as it doubles the predicted bounding box and mask with the target object. These identified challenges suggest further optimization of the training dataset so that the detection and segmentation for each epiphyte plant, parasitic plant, and host trees would be cleaner and not too complicated.

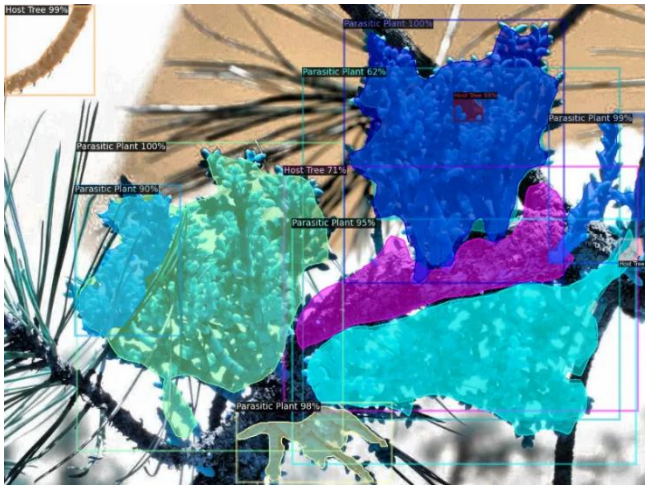


Figure 9. Overview of Object Detection and Segmentation of the Initial Output of Parasitic Plants on Host Trees

The Optimization of Mask R-CNN for Multiple Object Detection and Segmentation of Epiphyte Plants, Parasitic Plants, and Host Trees

The initial implementation of the Mask R-CNN model achieved great metrics and performance for the detection and segmentation of epiphyte plants, parasitic plants, and host trees. That initial implementation is highly significant as it would be the baseline for optimizing the model's metrics and performance based on the observed initial result. Further optimization strategies would be applied based on the observed problems and challenges after the implementation.

Training Setup for Optimization of Mask R-CNN

The first step of optimization is removing some of the image datasets that have too many occluded plants and a complex background. Reducing the annotations within each image has also been applied for a cleaner output of detection and segmentation within an object. Furthermore, reviewing the

number of instances per class has also been carefully monitored and balanced to avoid biases for each class.

Additionally, the dataset split was modified to an 80% training, 10% validation, and 10% testing ratio (80/10/10) to better evaluate model generalization and performance.

The continuation of the model optimization process has added various preprocessing and augmentation techniques for further improvement of the model. For preprocessing, resizing to 1024x1024 is added to avoid cutting or removing some of the significant objects within image datasets, such as epiphyte plants, parasitic plants, and host trees. Auto-Adjust Contrast with a value of Adaptive Equalization in Roboflow is also applied to automatically improve and balance how bright and dark the areas of an image and for clearer details of the objects. Augmentations are also extended, such as rotation, saturation, brightness, exposure, blur, and noise, to improve the robustness of the model's performance in terms of natural and complex environments.

Table 11. Total Number of Cleaned and Optimized Datasets

Split	**Number of Images**
Training	1335
Validation	56
Test	55
Total	1446

Table 12. Total Number of Instances per Class of Cleaned and Optimized Datasets

Class	Number of Instances
Epiphyte Plant	1868
Parasitic Plant	1556
Host Tree	2310
Total	5734

Table 13. Details of Preprocessing Used in Optimized Model Training

Preprocessing	Specification
Auto-Orient	Applied
Resizing	1024x1024
Auto-Adjust Contrast	Using Adaptive Equalization

Table 14. Details of Augmentations Used in Optimized Model Training

Augmentations	Specifications
Flip	Horizontal
Rotation	Between -15° and +15°
Brightness	Between -25% and +25%
Exposure	Between -5% and +5%
Saturation	Between -25% and +25%
Blur	Up to 2px
Noise	Up to 1% of pixels

Table 15. Details of Hyperparameters Used in Optimized Model Training

Hyperparameter	Specifications
Learning rate	(0.01, 0.005, 0.0025, 0.00125)
Weight Decay	0.0001
Batch Size	2
Epochs	10, 20, 30
Warmup Iterations	200
Number of Workers	2

The Result of the Implementation of Optimized Mask R-CNN

The implementation and application of the various techniques for optimization of the model's metrics and performance have achieved great improvement in mAP50 for bounding box and segmentation. The hyperparameters used from the initial implementation are retained except for epochs and the learning 0.02, as it crashes out during the initial implementation. For this optimization, we trained a little longer for 10, 20, and 30 epochs to give the model a chance to learn more, and the rest of the hyperparameters and model configurations are retained.

Table 16 presents the optimized Mask R-CNN, highlighting the significant impact of how the various optimization strategies greatly affect the improvement of the mAP50. After a careful evaluation of each result, the best and balanced results for bounding box and segmentation were achieved at a learning rate of 0.005 in 10 epochs with a backbone network of ResNet-101. This demonstrates that the optimization of hyperparameters and the refinement

of data preparation significantly enhance the overall metrics and performance of the model.

Table 16. Results from the Optimized Implementation in Mask R-CNN

Learning Rate	Epochs	RestNet-50		RestNet-101	
		Bbox mAP50	Segm mAP50	Bbox mAP50	Segm mAP50
0.01	10	83.482	75.104	82.783	75.064
	20	81.709	76.490	82.182	76.808
	30	80.019	76.221	79.220	74.491
0.005	10	82.666	74.080	83.007	76.746
	20	78.986	72.461	77.905	72.232
	30	78.784	73.388	79.065	73.563
0.0025	10	83.802	71.742	82.158	72.903
	20	79.938	69.811	81.015	75.396
	30	79.110	71.627	79.260	72.389
0.00125	10	82.624	69.682	82.241	71.392
	20	80.143	70.041	81.537	71.372
	30	79.462	68.474	79.946	72.665

Table 17 presents the performance results of the best optimized Mask R-CNN model. For the bounding box level, the model achieved a mAP score of 58.811 and an mAP50 score of 83.007. At the segmentation level, the model recorded a mAP of 37.082 and an mAP50 of 76.746. The precision, recall, and F1-score values were also evaluated at the optimal confidence threshold of 0.8 and at an IoU50 threshold. The bounding box level reached a precision of 0.6693, recall of 0.6849, and F1-score of 0.6768. For segmentation, the

precision was 0.5688, recall was 0.5820, and the F1-score was 0.5751. These values represent the final performance of the optimized model used in this study.

Table 17. Results of the Best Performing Optimized Implementation of Mask R-CNN

Metrics	Bounding Box Level	Segmentation Level
mAP	58.811	37.082
mAP50	83.007	76.746
Precision	0.6693	0.5688
Recall	0.6849	0.5820
F1-Score	0.6768	0.5751

Figure 10 presents the smooth decrease of training loss for the optimized Mask R-CNN, which indicates that the model is smoothly learning with the optimization techniques used. Even with a minimal increase in some of the iterations, the model found its way to decrease the loss and learn more.

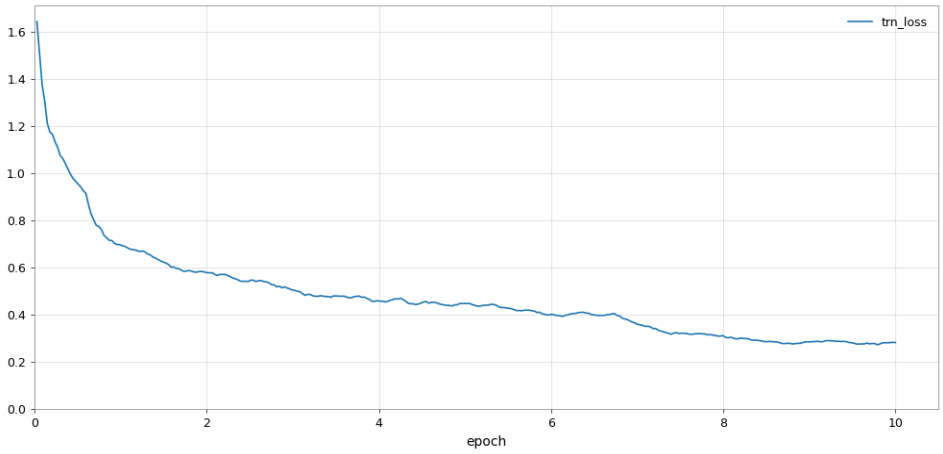


Figure 10. Training Loss Plot of Best Result in Optimized Mask R-CNN

Figure 11 presents the validation mAP of the optimized Mask R-CNN over the entire epochs. Based on the plot results, although the plot shows a very light increase each epoch, it doesn't suggest that the model fluctuates or overfits. A light or very low increase over time is still progress for the model.

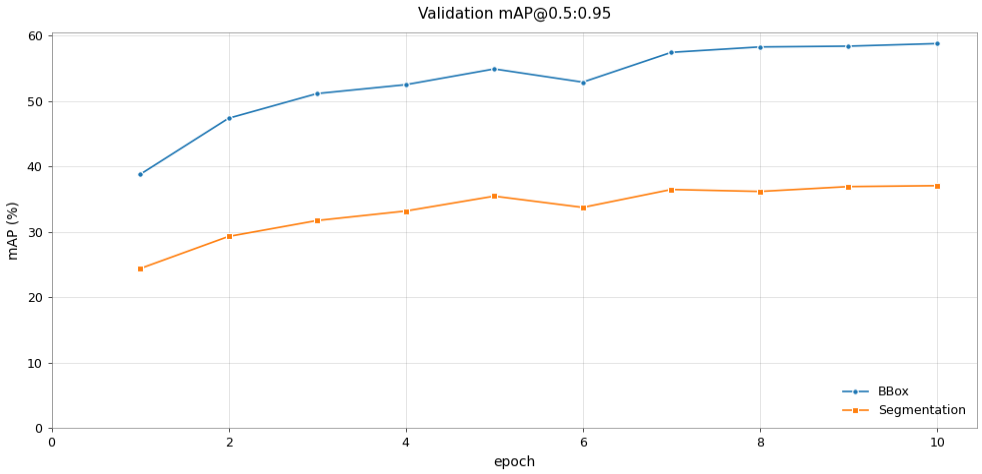


Figure 11. Validation mAP Plot of Best Result in Optimized Mask R-CNN

Figure 12 and Figure 13 presents the confusion matrix for both the bounding box and segmentation level for optimized Mask R-CNN. Based on the table result, both show a great detection and segmentation for each class, and minimal missed objects and confusion for each class for the bounding box and segmentation level compared to the initial implementation.

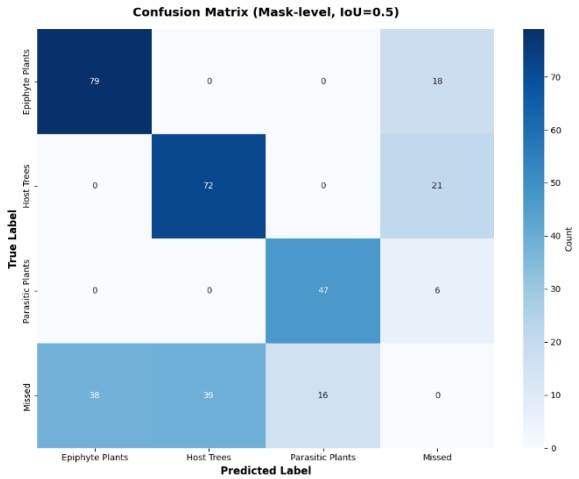


Figure 12. Confusion Matrix for bounding box level of Best Result in Optimized Mask R-CNN

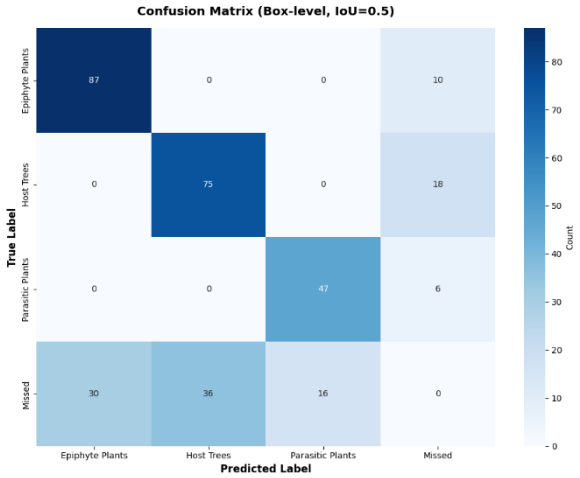


Figure 13. Confusion Matrix for the segmentation level of Best Result in Optimized Mask R-CNN

Figure 14 presents the performance of the optimized Mask R-CNN based on the various optimization techniques applied. Based on the performance of the model in epiphyte plants with their host tree, it successfully detects and segments, and even refines the mask over the target objects, especially the small objects that are not present in the initial implementation.

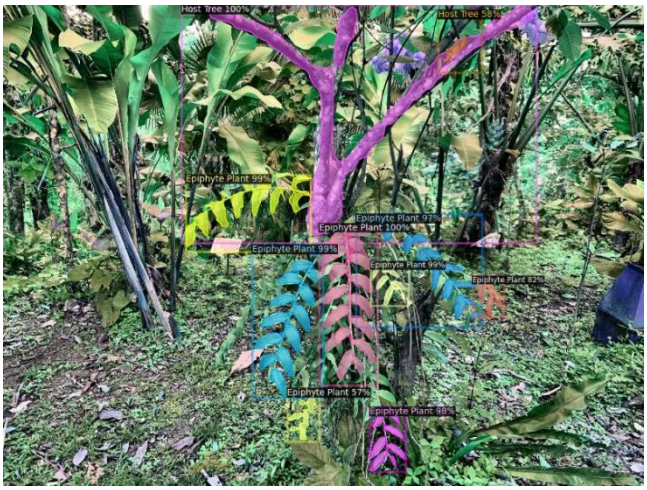


Figure 14. Overview of Object Detection and Segmentation of Optimized Output of Epiphyte Plants on Host Trees

Figure 15 presents the performance of the optimized Mask R-CNN on parasitic plants and their host tree. Based on the results, it is also successful in terms of detection and segmentation, and even refines the mask over the parasitic plants. Based on the observations, compared to the initial implementation result, the mask is now cleaner and clearer, which means it doesn't produce excess mask over the objects.

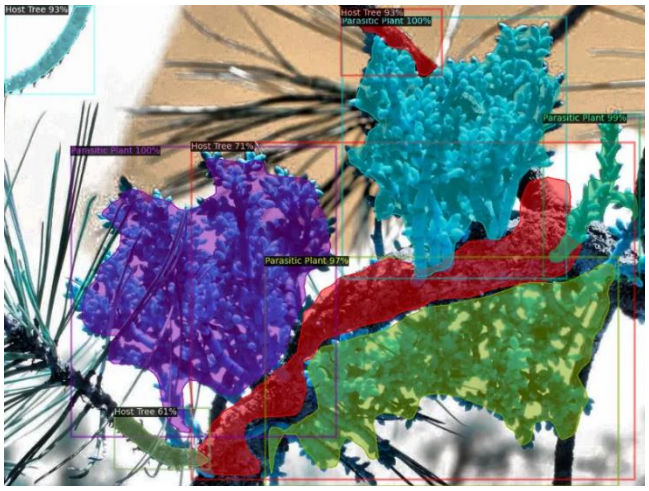


Figure 15. Overview of Object Detection and Segmentation of Optimized Output of Epiphyte Plants on Host Trees

Statistical Test

Table 18 presents the descriptive statistics of the Expert and Model scores based on one hundred (100) image samples. This presents the detailed comparison, highlighting any significant differences or alignments between the expert's evaluations and the model's predictions. This analysis is crucial for validating the effectiveness of the Mask R-CNN model in detecting and segmenting epiphyte/parasitic plants with their host tree and ensuring its applicability and practicality.

The expert obtained a mean score of 1.50 and a median of 1.5, indicating that the majority of expert evaluation were centered between the score of 1 and

2. The standard deviation (SD = 0.503) and standard error (SE = 0.050) suggest minimal variability among the expert evaluations, implying consistent scoring across the samples.

Meanwhile, the model achieved a slightly higher mean score of 1.56 with a median of 2.0, suggesting that the model tends to predict slightly a scores of 3 compared to the expert. The standard deviation (SD = 0.556) and standard error (SE = 0.056) indicate low variability, reflecting that the model's outputs are consistent and closely aligned with the expert's evaluations.

Table 18. Descriptive Statistics for Both Expert Evaluation and Optimized Mask R-CNN Prediction

Group	N	Mean	Median	SD	SE
Botanist Expert	100	1.50	1.5	0.503	0.050
Model	100	1.56	2.0	0.556	0.056

Table 19 presents the results of the paired samples t-test conducted to compare the expert evaluation scores and the optimized Mask R-CNN model prediction scores for identifying the plant types included in the dataset. This statistical test was performed to determine whether a statistically significant difference exists between human expert assessment and model-based predictions. Specifically, the test evaluates whether the observed variations between the two sets of scores are due to random chance or reflect a consistent performance gap.

The paired t-test produced a test statistic of t = –1.75, which represents the standardized magnitude of the difference between expert scores and model predictions. A negative value indicates that, on average, the model's scores are

slightly lower than the expert scores. The test was calculated with 99 degrees of freedom (df = 99), corresponding to the number of paired observations used in the analysis. These values collectively inform the inferential basis for determining whether the score differences are statistically meaningful.

Since the p-value (0.083) is greater than the 0.05 significance threshold, the results indicate that there is no statistically significant difference between the expert evaluations and the model predictions. The mean difference of –0.06, accompanied by a standard error of difference of 0.34, further clarifies the magnitude and stability of the observed difference. The negative mean difference indicates a slight tendency for the model to produce marginally lower scores than experts; however, the small absolute value demonstrates that this difference is trivial in practical terms. The relatively larger standard error, compared to the mean difference, reinforces the idea that the variations are unstable and may not indicate a consistent performance gap.

These conclusions are further supported by the 95% confidence interval for the mean difference, which ranges from –0.128 to 0.00804. Since this interval includes zero, it indicates that the true population difference may be nonexistent, thereby strengthening the finding that expert and model scores are statistically comparable. Confidence intervals offer a more detailed interpretation than p-values alone and indicate that the observed variability between the two sets of scores does not reflect a meaningful or substantial difference.

The analysis also includes the effect size, reported as Cohen’s d = –0.175, with a 95% confidence interval ranging from –0.372 to 0.0230. Cohen’s

d quantifies the practical magnitude of the difference between the two sets of scores. A value near zero, as observed here, indicates a very small effect size, suggesting that the difference—although measurable—is not practically meaningful. Since the confidence interval for the effect size also crosses zero, it further indicates that the estimated difference does not hold meaningful practical significance.

Overall, these findings demonstrate that the optimized Mask R-CNN model performs at a level statistically comparable to human experts, confirming its reliability and consistency in detecting and segmenting epiphyte plants or parasitic plants with their host trees. The minimal difference between expert and model evaluations indicates that the proposed model can replicate expert-level judgment with high consistency. This supports the practical applicability of the proposed deep learning–based detection model in real-world plant symbiosis analysis, where accurate and reliable automated detection is essential.

Table 19. Result of Paired Samples T-Test for Both Expert Evaluation and Optimized Mask R-CNN Prediction

T-Test	Statistic	df	p	Mean Difference	SE Difference
Paired T-test	-1.75	99	0.083	-0.06	0.34

95% Confidence Interval				95% Confidence Interval	
Lower	**Upper**		**Effect Size**	**Lower**	**Upper**
-0.128	0.00804	Cohen's d	-0.175	-0.372	0.0230

CHAPTER V

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

Summary

This study focused on the symbiotic relationship between epiphyte plants and parasitic plants on their host trees, as understanding how these plants depend on or affect each other is important for keeping forests healthy and supporting biodiversity. The dataset for epiphyte plants gathered at the Makiling Botanic Gardens (MBG) within the University of the Philippines Los Baños, part of the Mount Makiling Forest Reserve, which is recognized as an ASEAN Heritage Park and known for its rich biodiversity. However, parasitic plants were gathered from a public source since the target species was not seen in the study area. A total of 1446 images were collected and divided into training, validation, and testing sets, containing 5734 labeled object instances.

Mask R-CNN was used is this study capable of detecting and segmenting epiphyte plants, parasitic plants, and host trees for symbiotic relationship. Mask R-CNN was applied in model training using the Detectron2 framework in Python. It first used a ResNet-50 backbone with default hyperparameters as a baseline, then moved to a deeper ResNet-101 backbone with tuned hyperparameters to improve accuracy after initial testing. The training set helped the model learn from patterns of data, while the validation set monitored its performance, and the test set was reserved for final evaluation. Mask R-CNN performs instance segmentation and it was able to detect object boundaries even when plant parts were overlapping or hidden. During the first evaluation, the model achieved 74.598% mAP50 for object detection and 68.740% mAP50 for segmentation. After improving data

preparation, preprocessing, and model configuration, the performance increased to 83.007% mAP50 for detection and 76.746% mAP50 for segmentation. The model was also tested on real-world images and continued to show strong results even with different lighting, backgrounds, and angles.

Conclusions

This section presents a comprehensive summary of the study's conclusions. Each research objective is methodically examined to offer a clear interpretation of the results, ensuring that they align with the established research goals.

1. To implement the Mask R-CNN model for multiple object detection and segmentation of epiphyte plants, parasitic plants, and host trees using still image data related to plant symbiosis.

This study successfully implements the deep learning-based Mask R-CNN, capable of detecting and segmenting epiphyte plants, parasitic plants, and host trees for symbiotic relationship. From data preparation to hyperparameter tuning and adjusting model configuration, the model achieved 74.598% of mAP50 at the bounding box level and 68.740% mAP50 at the segmentation level, with a ResNet-101 backbone architecture for initial implementation and baseline of the study. This study proves that manually tuning the hyperparameter and configuration has a significant improvement in standard metrics.

2. To improve the detection and segmentation performance of a Mask R-CNN model for epiphyte plants and parasitic plants with their host trees by applying optimized data preparation and model tuning strategies

The application of various optimization strategies, from data cleaning, adding proper preprocessing and augmentations, hyperparameter tuning, and model configuration, the model achieved a huge improvement, from 74.598% to 83.007% of mAP50 at the bounding box level and from 68.740% to 76.746% of mAP50 at the segmentation level. This suggests that by carefully adjusting and optimizing the data preparations, hyperparameters, and model configurations, the model was successfully optimized with it.

3. To evaluate the performance of the Mask R-CNN model under real-world conditions using standard object detection and segmentation metrics within the study period.

The performance of the Mask R-CNN model was successfully tested using real-world image data to see how well it can detect and segment epiphyte plants, parasitic plants, and host trees in natural environments. The model showed consistent and reliable results, maintaining high performance accuracy even with varying lighting, backgrounds, and plant positions. The use of standard evaluation metrics such as mAP50 for both detection and segmentation confirmed that the model performs effectively beyond controlled testing conditions. This signifies the Mask R-CNN model can be applied in actual field scenarios to help in identifying and analyzing plant symbiotic relationships.

Recommendations

The study's results and findings have led to the suggestion of several number of recommendations for future researchers, further improvement, and future development.

1. Future researchers are encouraged to adopt the structured experimental approach used in this study to further enhance and optimize the Mask R-CNN model. This includes careful data preparation, application of data augmentation techniques, and a systematic training process that explores various learning rates, epoch settings, and backbone architectures. The proposed experimental framework offers a clear and organized workflow that promotes consistency and reliability in model evaluation. By following this approach, future studies can conduct more efficient experiments and achieve more accurate and reproducible performance outcomes.
2. Future researchers are encouraged to extend the study by including a classification model that can categorize between different types or species of epiphyte plants and parasitic plants. Integrating both detection and classification tasks within a single framework could contribute to a more comprehensive understanding of symbiotic relationships in various forest .
3. Future researchers should also consider expanding the dataset of this study to include a wider variety of species under real world conditions using a high quality camera with different lenses size that can capture in various range in high resolution. Gathering more diverse examples of epiphyte plants and parasitic plants will improve the model's generalization ability and reduce overfitting in training the data.
4. Future researchers needs to spend more time in annotation process due to the difficulties and time consuming in annotating plants and trees structure. Additionally, applying image preprocessing and proper augmentation techniques may help minimize errors and increase the accuracy of detection and segmentation in real-world scenarios. This will allow the

model to avoid learning incorrect patterns in the dataset and improve its ability to accurately detect and segment.

5. Training deep learning model with large dataset require high computational power in this study, so future researchers are encourage to use computer or laptop with high specifications. Using devices with powerful graphic cards (GPU) and faster processor will help reduce training time.
6. Future researchers are encouraged to apply the implemented model in real time object detection and recognition task. This process could lead to the development of intelligent tools for quick assessment of host trees and their associated plant species in natural habitats.

REFERENCES

Abd Alhadi, A. R. S., & Naji, R. K. (2025). The role of amensalism and parasitism on the dynamics of three-species ecological system. *Results in Control and Optimization, 19*, 100571.

Afzaal, H., Awais, M., Riaz, A., Qureshi, M. A., & Usman, M. (2021). An instance segmentation model for strawberry diseases based on Mask R-CNN. *Sensors, 21*(19), 6565.

Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A next-generation hyperparameter optimization framework. *arXiv*.

Ali, S. H. (2025). *Parasitic plants: Their distribution, economic damages and methods of control*. ResearchGate.

Amit, K., Jha, R., & Kumar, R. (2021). Object detection. In *Encyclopedia of Renewable and Sustainable Materials* (pp. 660–674). Springer.

Aralikatti, R. K., Deshpande, A., & Koti, A. (2020). Comparative analysis of object detection algorithms. *Journal of Physics: Conference Series, 1706*(1), 012149.

Aslam, N., Javed, M. Y., Bibi, A., & Noor, R. M. (2020). Object detection and localization in natural scenes through single-step and two-step models. *2020 3rd International Conference on Computing, Mathematics and Engineering Technologies (iCoMET)*, 1–6.

Barcelona, J. F., Pelser, P. B., Balete, D. S., & Co, L. L. (2006). Taxonomy, ecology, and conservation status of Philippine Rafflesia species. *Blumea, 51*(1), 77–93.

Barkman, T. J., Lim, S. H., Salleh, K. M., & Nais, J. (2004). Mitochondrial DNA sequences reveal the photosynthetic relatives of *Rafflesia*, the world's largest flower. *Proceedings of the National Academy of Sciences, 101*(3), 787–792.

Barthlott, W., Schmit-Neuerburg, V., Nieder, J., & Engwald, S. (2001). Diversity and abundance of vascular epiphytes: A comparison of secondary vegetation and primary montane rain forest in the Venezuelan Andes. *Plant Ecology, 152*(2), 145–156.

Beery, S., Morris, D., & Yang, S. (2018). Recognition in terra incognita. In *European Conference on Computer Vision*.

Bergstra, J., & Bengio, Y. (2012). Random search for hyper-parameter optimization. *Journal of Machine Learning Research, 13*(1), 281–305.

Bhadana, D., & Kurunthachalam, A. (2025). Deep learning approach for seed disease prediction using VGG16 with transfer learning and Optuna optimization. *International Journal of Advanced Research in Science, Communication and Technology*, 604–611.

Bi, Y., Liu, H., Wang, W., & Zhao, R. (2023). Comparing one-stage and two-stage learning strategy in object detection. *Asia Conference on Engineering*.

Bochkovskiy, A., Wang, C. Y., & Liao, H. Y. M. (2020). YOLOv4: Optimal speed and accuracy of object detection. *arXiv preprint arXiv:2004.10934*.

Bottan, L., Rodríguez-Souilla, J., Benitez, J., Barrera, M. D., Cellini, J. M., Martínez Pastur, G. J., & Lencinas, M. V. (2025). Effects of variable retention on the interaction between tree regeneration and understory plants at different

regeneration heights over 15 years in *Nothofagus pumilio* forests. *Forest Ecology and Management, 586*, 122725.

Bronstein, J. L. (2009). II.11 Mutualism and symbiosis. In S. A. Levin, S. R. Carpenter, H. C. J. Godfray, A. P. Kinzig, M. Loreau, J. B. Losos, B. Walker, & D. S. Wilcove (Eds.), *The Princeton Guide to Ecology* (pp. 233–238). Princeton University Press.

Brooks, T. M., Mittermeier, R. A., da Fonseca, G. A. B., Gerlach, J., Hoffmann, M., Lamoreux, J. F., … Rodrigues, A. S. L. (2002). Habitat loss and extinction in the hotspots of biodiversity. *Conservation Biology, 16*(4), 909–923.

Brundrett, M. C. (2009). Mycorrhizal associations and other means of nutrition of vascular plants: Understanding the global diversity of host plants by resolving conflicting information and developing reliable means of diagnosis. *Plant and Soil, 320*(1), 37–77.

Bruna, E. M., & Ribeiro, M. B. N. (2005). The compensatory responses of an understory herb to experimental damage are habitat-dependent. *American Journal of Botany, 92*(12), 2101–2106.

Chen, J., et al. (2021). Deep learning for plant identification using mixed model approaches. *Computers and Electronics in Agriculture, 185*, 106112.

Chen, X., Fan, H., Girshick, R., & He, K. (2019). Improved baselines with momentum contrastive learning. *arXiv preprint arXiv:2003.04297*.

Chen, Y., Xu, Y., Zhang, X., & Zhang, L. (2021). A real-time object detection approach for industrial scenes based on an improved YOLOv4. *Electronics, 10*(4), 372.

Chowdhury, M., Reza, M. N., Jin, H., Islam, S., Lee, G., & Chung, S. (2024b). Defective pennywort leaf detection using machine vision and Mask R-CNN model. *Agronomy, 14*(10), 2313.

Cruz, R. V., Pulhin, J. M., & Lasco, R. D. (2019). Climate change impacts and adaptation in Philippine forests. *Asian Journal of Agriculture and Development, 16*(1), 1–15.

Da Cruz, U. V. G., Nogueira, T. D. C., Da Cruz, G., Junior, Vinhal, C. D. N., Ullmann, M. R. D., Carvalho, C. H. R., & De Oliveira Santana, D. (2025). Deep learning for detection of foliar diseases in soybeans based on the Mask R-CNN model. *Revista de Gestão Social e Ambiental, 19*(1), e011121.

Delight, M., & Karunakaran, S. (2021). Deep learning based object detection using Mask R-CNN. In *2021 International Conference on Artificial Intelligence and Machine Vision (AIMV)* (pp. 243–247).

Du, S., Zhang, Z., Huang, L., & Zhang, X. (2020). Overview of two-stage object detection algorithms. *Journal of Physics: Conference Series, 1544*(1), 012033.

Dyrmann, M., Karstoft, H., & Midtiby, H. S. (2016). Plant species classification using deep convolutional neural network. *Biosystems Engineering, 151*, 72–80.

Egorov, A. D., & Reznik, M. S. (2021). Selection of hyperparameters and data augmentation method for diverse backbone models Mask R-CNN. Proceedings of the 2021 IV International Conference on Control in Technical Systems (CTS), 249–251. IEEE.

Ganatra, S. (2021). Comparative study of YOLO and SSD for real-time object detection. In *2021 6th International Conference on Communication and Electronics Systems (ICCES)* (pp. 1243–1247).

Karl, F., Pielok, T., Moosbauer, J., Pfisterer, F., Coors, S., Binder, M., Schneider, L., Thomas, J., Richter, J., Lang, M., Garrido-Merchán, E. C., Branke, J., & Bischl, B. (2022). *Multi-Objective Hyperparameter Optimization — an overview. arXiv (Cornell University).*

Garcia-Garcia, A., Orts-Escolano, S., Oprea, S., Villena-Martinez, V., & Garcia-Rodriguez, J. (2017). A review on deep learning techniques applied to semantic segmentation. *arXiv preprint arXiv:1704.06857*.

Gentry, A. H., & Dodson, C. H. (1987). Diversity and biogeography of neotropical vascular epiphytes. *Annals of the Missouri Botanical Garden, 74*(2), 205–233.

Groom, M., Meffe, G. K., & Carroll, C. R. (2006). *Principles of conservation biology*. Sinauer Associates.

He, K., Gkioxari, G., Dollár, P., & Girshick, R. (2017a). Mask R-CNN. In *Proceedings of the IEEE International Conference on Computer Vision* (pp. 2961–2969).

He, K., Gkioxari, G., Dollár, P., & Girshick, R. (2017b). Mask R-CNN. *arXiv preprint arXiv:1703.06870*.

Hebbar, P. V., & Pullela, S. (2023). Enhanced detection with YOLOv5: A practical approach. In *2023 International Conference on Computer Communication and Informatics (ICCCI)* (pp. 1–6).

Hoang, V.-H., Lee, J. W., & Park, C.-S. (2025). Enhancing fire detection with YOLO models: A Bayesian hyperparameter tuning approach. *Computers, Materials & Continua, 83*(3), 4097–4116.

Hossain, M. A., Qiu, X., & Li, C. (2019). BackNet: An enhanced backbone network for accurate detection of objects with large scale variations. In *Advances in Intelligent Systems and Computing* (pp. 469–478). Springer.

Ibáñez, I., McPherson, M. R., Upchurch, R. A., & Zak, D. R. (2025). *Mycorrhizal fungi influence on mature tree growth: Stronger in high-nitrogen soils for an EMF-associated tree and in low-nitrogen soils for two AMF-associated trees*. *Plant-Environment Interactions, 6*(3), e70055.

Islam, S., Reza, M. N., Chowdhury, M., Ahmed, S., Lee, K., Ali, M., Cho, Y. J., Noh, D. H., & Chung, S. (2024). Detection and segmentation of lettuce seedlings from seedling-growing tray imagery using an improved Mask R-CNN method. *Smart Agricultural Technology, 8*, 100455.

Jalled, F., & Voronkov, A. (2016). A survey of object detection methods. *arXiv preprint arXiv:1611.07791*.

Jiao, L., Zhang, F., Liu, F., Yang, S., Li, L., Feng, Z., & Qu, R. (2019). A survey of deep learning-based object detection. *IEEE Access, 7*, 128837–128868.

Jocher, G., et al. (2023). *YOLOv8 Documentation*. Ultralytics.

Johnson, N. C., Graham, J. H., & Smith, F. A. (2010). Functioning of mycorrhizal associations along the mutualism–parasitism continuum. *New Phytologist*.

Kaya, S., Kiraz, B., & Çamurcu, A. Y. (2024). Hyperparameter optimization in deep learning-based object detection of branching and endpoints on 2D brain vessel

images. *2024 Innovations in Intelligent Systems and Applications Conference (ASYU)*, Ankara, Turkiye, 1–6. IEEE.

Khan, A., Ali, M., Khan, M., Shah, F., & Kim, K. (2023). YOLO-based real-time multiple object detection and tracking system for smart surveillance. *Electronics, 12*(13), 2768.

Kong, T., Sun, F., Liu, H., Jiang, Y., & Shi, J. (2019). FoveaBox: Beyond anchor-based object detector. *IEEE Transactions on Image Processing, 29*, 7389–7398.

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems* (Vol. 25, pp. 1097–1105).

Kussul, N., Lavreniuk, M., Skakun, S., & Shelestov, A. (2017). Deep learning classification of land cover and crop types using remote sensing data. *IEEE Geoscience and Remote Sensing Letters, 14*(5), 778–782.

Li, L., Zhang, H., & Liu, C. (2020). AutoML for object detection: Optimizing architecture and hyperparameters. *International Journal of Computer Vision, 128*, 3230–3246.

Lin, T. Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., … Zitnick, C. L. (2014). Microsoft COCO: Common objects in context. In *European Conference on Computer Vision* (pp. 740–755).

Liu, Y., Wang, G., Wang, H., & Liu, X. (2022). Deep learning for pest detection in agriculture: A review. *Computers and Electronics in Agriculture, 193*, 106743.

Long, T., Zhao, Y., Zhang, Y., & Wang, H. (2020). A survey of two-stage object detection algorithms with different backbones. In *2020 IEEE International*

Conference on Artificial Intelligence and Computer Applications (ICAICA) (pp. 124–129).

Mack, R. N., Simberloff, D., Lonsdale, W. M., Evans, H., Clout, M., & Bazzaz, F. A. (2000). Biotic invasions: Causes, epidemiology, global consequences, and control. *Ecological Applications, 10*(3), 689–710.

Mathis, K. A., & Bronstein, J. L. (2020). Our current understanding of commensalism. *Annual Review of Ecology, Evolution, and Systematics, 51*, 167–189.

Murel, N., & Kavlakoglu, E. (2024). What is object detection? *IBM*.

Multi detection and segmentation coconut shell for charcoal briquette using Mask R-CNN. (2023, July 26). *IEEE Conference Publication*.

Nadkarni, N. M. (1994). Diversity of species and interactions in the upper tree canopy of forest ecosystems. *American Zoologist, 34*(1), 70–78.

Nadkarni, N. M., & Matelson, T. J. (1992). Biomass and nutrient dynamics of epiphytes and canopy trees in a Neotropical cloud forest. *Ecology, 73*(1), 170–189.

Nimis, P. L., Scheidegger, C., & Wolseley, P. A. (2002). *Monitoring with lichens—Monitoring lichens*. Springer.

Panda, S. (2019). Object detection and tracking using Faster R-CNN. *International Journal of Recent Technology and Engineering, 8*(3), 218–221.

Pelser, P. B., Barcelona, J. F., & Nickrent, D. L. (2011). *Co's Digital Flora of the Philippines*.

Posa, M. R. C., Diesmos, A. C., Sodhi, N. S., & Brooks, T. M. (2008). Hope for threatened tropical biodiversity: Lessons from the Philippines. *Bioscience, 58*(3), 231–240.

Qian, Y., Ma, S., Sun, J., & Liu, B. (2018). Object detection using deep convolutional neural networks. In *2018 Chinese Automation Congress (CAC)* (pp. 568–573).

Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). You only look once: Unified, real-time object detection. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 779–788).

Redmon, J., & Farhadi, A. (2018). YOLOv3: An incremental improvement. *arXiv preprint arXiv:1804.02767*.

Ren, S., He, K., Girshick, R., & Sun, J. (2015). Faster R-CNN: Towards real-time object detection with region proposal networks. In *Advances in Neural Information Processing Systems* (Vol. 28, pp. 91–99).

Ren, S., He, K., Girshick, R., & Sun, J. (2016). Faster R-CNN: Towards real-time object detection with region proposal networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence, 39*(6), 1137–1149.

Ramos, H. M., & Malabrigo, P. L. Jr. (2020). Indigenous knowledge systems and practices on forest conservation in the Philippines. *Journal of Nature Studies, 19*(1), 37–52.

Rodriguez, R. J., White Jr, J. F., Arnold, A. E., & Redman, R. S. (2009). Fungal endophytes: Diversity and functional roles. *New Phytologist, 182*(2), 314–330.

Sa, I., Ge, Z., Dayoub, F., Upcroft, B., Perez, T., & McCool, C. (2017). DeepFruits: A fruit detection system using deep neural networks. *Sensors, 16*(8), 1222.

Saddique, M. A., Usman, M., & Siddiqui, M. S. (2020). YOLO-based object detection: A survey. In *2020 International Conference on Computing, Electronic and Electrical Engineering (ICE Cube)* (pp. 1–6).

Selosse, M. A., Richard, F., He, X., & Simard, S. W. (2014). Mycorrhizal networks: Des liaisons dangereuses? *Trends in Ecology & Evolution, 21*(11), 621–628.

Shanthi, N., Rampradheep, G. S., Muthuraja, M., KalaiSelvan, K., Madhumita, C., & Malathi, S. (2025). Optimizing construction and demolition waste segmentation using advanced deep learning techniques (Semantic vs instance segmentation). *2025 International Conference on Emerging Smart Computing and Informatics (ESCI)*, Pune, India, 1–6. IEEE.

Sharma, P., & Thakur, N. (2017). A survey and analysis of object detection techniques. *International Journal of Computational Vision and Robotics, 7*(2), 196–217.

Shorten, C., & Khoshgoftaar, T. M. (2019). A survey on image data augmentation for deep learning. *Journal of Big Data, 6*(1), 60.

Singh, Geetika & Kumar, Dr. (2025). Symbiotic Relationships: Plants and Their Partners.

Siricharoen, W., Pusittigul, N., & Piriyajitakonkiti, C. (2023). Fruit maturity grading framework for small dataset using single image multi-object sampling and Mask R-CNN. *Engineering Reports, 3*(5), 100958.

Smith, S. E., & Read, D. J. (2008). *Mycorrhizal symbiosis* (3rd ed.). Academic Press.

Snoek, J., Larochelle, H., & Adams, R. P. (2012). Practical Bayesian optimization of machine learning algorithms. In *Advances in Neural Information Processing Systems* (Vol. 25, pp. 2951–2968).

Sodhi, N. S., Koh, L. P., Brook, B. W., & Ng, P. K. L. (2010). Southeast Asian biodiversity: An impending disaster. *Trends in Ecology & Evolution, 19*(12), 654–660.

Tedersoo, L., Bahram, M., Toots, M., et al. (2020). Global diversity and geography of soil fungi. *Science, 346*(6213), 1256688.

Turner, W., Spector, S., Gardiner, N., Fladeland, M., Sterling, E., & Steininger, M. (2003). Remote sensing for biodiversity science and conservation. *Trends in Ecology & Evolution, 18*(6), 306–314.

Uijlings, J. R. R., van de Sande, K. E. A., Gevers, T., & Smeulders, A. W. M. (2013). Selective search for object recognition. *International Journal of Computer Vision*.

University of Southampton. (2020, January 1). Improving object detection performance by lightweight approaches. *ePrints Soton*.

Van Der Heijden, M. G. A., Martin, F. M., Selosse, M. A., & Sanders, I. R. (2015). Mycorrhizal ecology and evolution: The past, the present, and the future. *New Phytologist, 205*(4), 1406–1423.

Varghese, N., & Sambath, S. (2023). Comparative study on object detection techniques using YOLOv5 and Faster R-CNN. In *2023 5th International Conference on Smart Systems and Inventive Technology (ICSSIT)* (pp. 105–110).

Wäldchen, J., & Mäder, P. (2018). Plant species identification using computer vision techniques: A systematic literature review. *Archives of Computational Methods in Engineering, 25*(2), 507–543.

Wang, C. Y., Bochkovskiy, A., & Liao, H. Y. M. (2022). YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. *arXiv preprint arXiv:2207.02696*.

Watson, D. M. (2001). Mistletoe: A keystone resource in forests and woodlands worldwide. *Annual Review of Ecology and Systematics, 32*, 219–249.

Weinstein, B. G. (2018). Scene-specific convolutional neural networks for video-based biodiversity detection. *Methods in Ecology and Evolution*.

Wu, Y., Wang, X., Chen, X., & Zhao, Z. (2020). Object detection based on RGC Mask R-CNN. *IET Image Processing, 14*(2), 284–290.

Xie, C., Wang, R., Zhang, J., Chen, P., Dong, W., Li, R., & Chen, T. (2018). A deep-learning-based real-time detector for grape leaf diseases using improved convolutional neural networks. *Frontiers in Plant Science*.

Xing, M., Hassan, M. F., & Yusof, R. (2023). Deep learning-based model for multi-object image detection. *Universiti Teknologi MARA Institutional Repository*.

Zhang, T. (2024). Overview of object detection based on deep learning. In *Asia Conference on Engineering*.

Zhang, X., Li, Y., & Liu, Y. (2017). Multiple objects detection based on improved Faster R-CNN. In *Proceedings of the 2017 International Conference on Robots & Intelligent System (ICRIS)* (pp. 38–42).

Zhang, Y., & Cloutier, G. (2022). Review on one-stage object detection based on deep learning. *ResearchGate*.

Zhang, Z., Xu, Y., Huang, Y., & Yu, H. (2021). Efficient object detection based on improved YOLOv4 algorithm. In *2021 5th International Conference on Computer Science and Artificial Intelligence (CSAI)* (pp. 283–289).

Zhao, Z. Q., Zheng, P., Xu, S. T., & Wu, X. (2019). Object detection with deep learning: A review. *IEEE Transactions on Neural Networks and Learning Systems, 30*(11), 3212–3232.

Zhou, S., Wang, G., & Lu, H. (2019). Robust Faster R-CNN: Increasing robustness to occlusions and multi-scale objects. In *Pattern Recognition and Machine Intelligence* (pp. 377–387). Springer.

Zhou, T., Han, G., Zhang, Y., & Guo, Y. (2020). A comparative study on deep learning models for plant species classification. *Neurocomputing, 383*, 271–279.

Zotz, G. (2013). *The biology of epiphytes*. Springer.

Zotz, G., & Hietz, P. (2001). The physiological ecology of vascular epiphytes: Current knowledge, open questions. *Journal of Experimental Botany, 52*(364), 2067–2078.

APPENDIX A

Approval Forms

CERTIFICATE OF APPROVAL

This is to certify that after having satisfactorily completed all the requirements of the thesis project entitled:

“DEEP LEARNING-BASED MULTI-OBJECT DETECTION OF EPIPHYTES AND PARASITIC PLANTS ON HOST TREES USING OPTIMIZED MASK R-CNN FOR SYMBIOTIC RELATIONSHIP”

ALEJANDREI APOLO M. DURAN, CARLOS MIGUEL Z. ROSALES have been granted permission to proceed with their proposal presentation.

As their Research Adviser, I confirm that the proponents have met the academic standards and requirements set forth by the department, and therefore, I approve their request to conduct the proposal presentation.

CRISANTO F. GULAY
Thesis Project Adviser

APPROVAL SHEET

The Thesis entitled **"DEEP LEARNING-BASED MULTI-OBJECT DETECTION OF EPIPHYTES AND PARASITIC PLANTS ON HOST TREES USING OPTIMIZED MASK R-CNN FOR SYMBIOTIC RELATIONSHIP"** prepared and submitted by Alejandrei Apolo M. Duran and Carlos Miguel Z. Rosales in partial fulfillment of the requirements for the degree, Bachelor of Science in Computer Science specialized in Intelligent Systems is at this moment recommended for approval and acceptance.

CRISANTO F. GULAY
Thesis Project Adviser

APPENDIX B

Certificate of Acceptance

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna Los Baños Campus
College of Computer Studies

July 1, 2025

Dear Director,

We are Bachelor of Science in Computer Science students from LSPU – Los Baños Campus, currently conducting our study entitled "DEEP LEARNING-BASED MULTIOBJECT DETECTION OF EPIPHYTES AND PARASITIC PLANTS ON HOST TREES USING OPTIMIZED MASK R-CNN FOR SYMBIOTIC RELATIONSHIP." Our research focuses on plant symbiosis, specifically on detecting epiphytes and parasitic plants on trees to determine its symbiotic relationship using a deep learning-based model for multiple object detection.

In line with this, we would like to respectfully request your assistance and approval to take photographs of any species of epiphytes and parasitic plants growing on trees within Makiling Botanic Garden. These photos will serve as image data for training and testing the deep learning model in our study. The images will be used strictly for academic purposes and will not be shared or published outside the scope of our thesis.

Currently, chapters 1 to 3 of our study are still in process for a few revisions, and being able to collect image data during this stage would greatly support and strengthen the foundation of our research.

We hope for your kind consideration and support, as this will be a valuable contribution to the success of our research. Thank you very much for your time.

With sincere appreciation,

Alejandrei Apolo M. Duran

Carlos Miguel Z. Rosales

Researchers, BS Computer Science
Email: alejandrei1duran18@gmail.com
Contact: 0962 336 7401

Noted by:

CRISANTO F. GULAY
Thesis Adviser

APPENDIX C

Relevant Source Code

Training Preparation

```
!pip install -U torch torchvision
!pip install git+https://github.com/facebookresearch/detectron2.git
!pip install -U pycocotools
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
DATASET_PATH = '/content/drive/MyDrive/dataset_17.coco-segmentation'
```

Dataset Registration

```
from detectron2.data.datasets import register_coco_instances
import os

train_images = os.path.join(DATASET_PATH, "train")
val_images = os.path.join(DATASET_PATH, "valid")

train_json = os.path.join(train_images, "_annotations_cleaned.coco.json")
val_json = os.path.join(val_images, "_annotations_cleaned.coco.json")

register_coco_instances("my_train", {}, train_json, train_images)
register_coco_instances("my_val", {}, val_json, val_images)

print("Dataset registered successfully!")
```

Model Configuration

```
from detectron2.engine import DefaultTrainer
from detectron2.config import get_cfg
from detectron2 import model_zoo
import os

cfg = get_cfg()
cfg.merge_from_file(model_zoo.get_config_file("COCO-InstanceSegmentation/mask_rcnn_R_50_FPN_3x.yaml"))

cfg.DATASETS.TRAIN = ("my_train",)
cfg.DATASETS.TEST = ("my_val",)

cfg.DATALOADER.NUM_WORKERS = 2

cfg.MODEL.WEIGHTS = model_zoo.get_checkpoint_url("COCO-InstanceSegmentation/mask_rcnn_R_50_FPN_3x.yaml")

cfg.SOLVER.IMS_PER_BATCH = 2  # Batch Size
cfg.SOLVER.BASE_LR = 0.01  # Learning rate
cfg.SOLVER.WEIGHT_DECAY = 0.0001 # Weight Decay
cfg.SOLVER.MAX_ITER = 6680    # Iterations

cfg.SOLVER.WARMUP_ITERS = 200
cfg.SOLVER.STEPS = (4453, 6012)

cfg.MODEL.ROI_HEADS.BATCH_SIZE_PER_IMAGE = 512


cfg.MODEL.ROI_HEADS.NUM_CLASSES = 3

cfg.OUTPUT_DIR = "/content/output"
os.makedirs(cfg.OUTPUT_DIR, exist_ok=True)

cfg.TEST.EVAL_PERIOD = 668
```

Trainer Setup and Evaluator Setup

```
from detectron2.engine import DefaultTrainer
from detectron2.evaluation import COCOEvaluator, inference_on_dataset
from detectron2.data import build_detection_test_loader

class TrainerWithEval(DefaultTrainer):
    @classmethod
    def build_evaluator(cls, cfg, dataset_name, output_folder=None):
        if output_folder is None:
            output_folder = os.path.join(cfg.OUTPUT_DIR, "inference")
        return COCOEvaluator(dataset_name, cfg, False, output_folder)
```

Model Training

```
from detectron2.evaluation import COCOEvaluator, inference_on_dataset
from detectron2.data import build_detection_test_loader

evaluator = COCOEvaluator("my_val", output_dir="./output")
val_loader = build_detection_test_loader(cfg, "my_val")
inference_on_dataset(trainer.model, val_loader, evaluator)
```

Model Evaluation

```
trainer = TrainerWithEval(cfg)
trainer.resume_or_load(resume=False)
trainer.train()
```

APPENDIX D

Evaluation Tool

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna Los Baños Campus
College of Computer Studies

DEEP LEARNING-BASED MULTI-OBJECT DETECTION OF EPIPHYTES AND PARASITIC PLANTS ON HOST TREES USING OPTIMIZED MASK R-CNN FOR SYMBIOTIC RELATIONSHIP

Alejandrei Apolo M. Duran and Carlos Miguel Z. Rosales
Researchers

The primary goal of our research is to develop a deep learning model that is relevant to the symbiotic relationship. This is achieved by developing an advanced deep learning model using Mask R-CNN to detect and segment patterns of plants and host tree.

This questionnaire, referred to in our study as the Evaluation Tool, is designed to help us evaluate the dataset, which consists of epiphyte plants and parasitic plants on host tree. Your honest assessment and expert insights are essential for the validation and refinement of our model.

Validation Procedure

The goal of this validation is to determine whether the dataset used in developing deep learning model is reliable. The validation process will involve the following steps:

1. **Data Provision:** Our research team will provide you with a minimum of 100 images. Each image will present an instance of epiphyte plants or parasitic plants on their host tree.

2. **Expert Validation:** You, As the expert validator, your role is to observe and confirm if our dataset represents epiphyte plants and parasitic plants on their host tree, which is relevant to the symbiotic relationship.

3. **Data Scoring:** Following your review, a scoring table will be provided for you to mark the dataset on a per-image basis:

   A score of "**1**" will be assigned if the image is confirmed as a type of epiphyte plant on a host tree.

   A score of "**2**" will be assigned if the image is confirmed as a type of parasitic plant on a host tree.

   A score of "**3**" will be assigned if the image is confirmed neither as a type of epiphyte plant nor parasitic plant on a host tree

This validation will ultimately establish the reliability of the dataset in the developed deep learning model

Please be assured that all your responses will be kept strictly confidential and used solely for academic and research purposes only.

VALIDATION TABLE

Images	Score
Image 1	1
Image 2	1
Image 3	1
Image 4	1
Image 5	1
Image 6	1
Image 7	1
Image 8	1
Image 9	1
Image 10	1
Image 11	1
Image 12	1
Image 13	1
Image 14	1
Image 15	1
Image 16	1
Image 17	1
Image 18	1
Image 19	1
Image 20	1
Image 21	1
Image 22	1
Image 23	1
Image 24	1
Image 25	1
Image 26	1
Image 27	1
Image 28	1
Image 29	1
Image 30	1

Images	Score
Image 51	2
Image 52	2
Image 53	2
Image 54	2
Image 55	2
Image 56	2
Image 57	2
Image 58	2
Image 59	2
Image 60	2
Image 61	2
Image 62	2
Image 63	2
Image 64	2
Image 65	2
Image 66	2
Image 67	2
Image 68	2
Image 69	2
Image 70	2
Image 71	2
Image 72	2
Image 73	2
Image 74	2
Image 75	2
Image 76	2
Image 77	2
Image 78	2
Image 79	2
Image 80	2

Image 31	1
Image 32	1
Image 33	1
Image 34	1
Image 35	1
Image 36	1
Image 37	1
Image 38	1
Image 39	1
Image 40	1
Image 41	1
Image 42	1
Image 43	1
Image 44	1
Image 45	1
Image 46	1
Image 47	1
Image 48	1
Image 49	1
Image 50	1

Image 81	2
Image 82	2
Image 83	2
Image 84	2
Image 85	2
Image 86	2
Image 87	2
Image 88	2
Image 89	2
Image 90	2
Image 91	2
Image 92	2
Image 93	2
Image 94	2
Image 95	2
Image 96	2
Image 97	2
Image 98	2
Image 99	2
Image 100	2



ANALYN C. MALAYBA

Signature over Printed Name

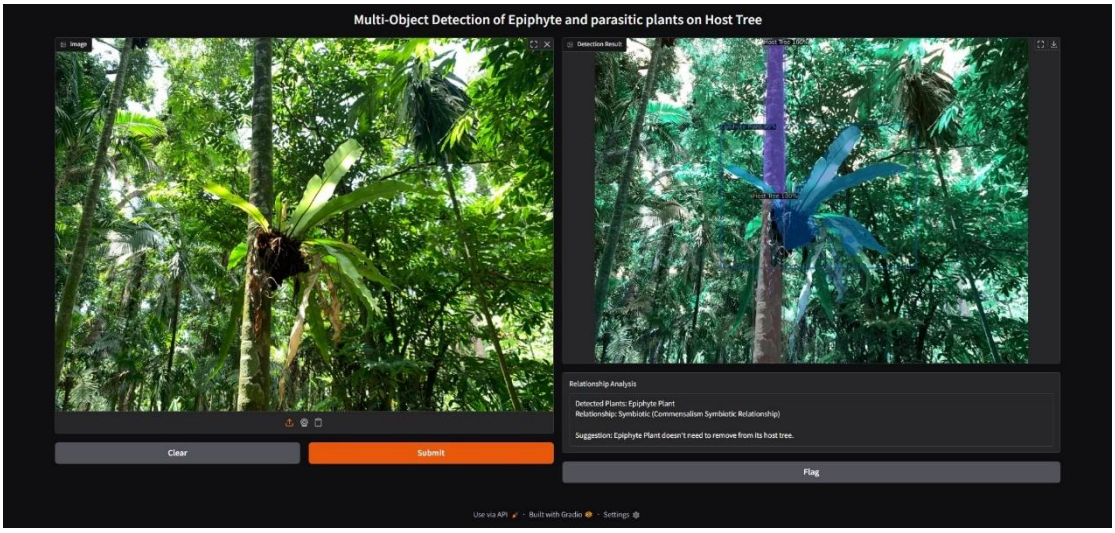
Licence No.: 006954

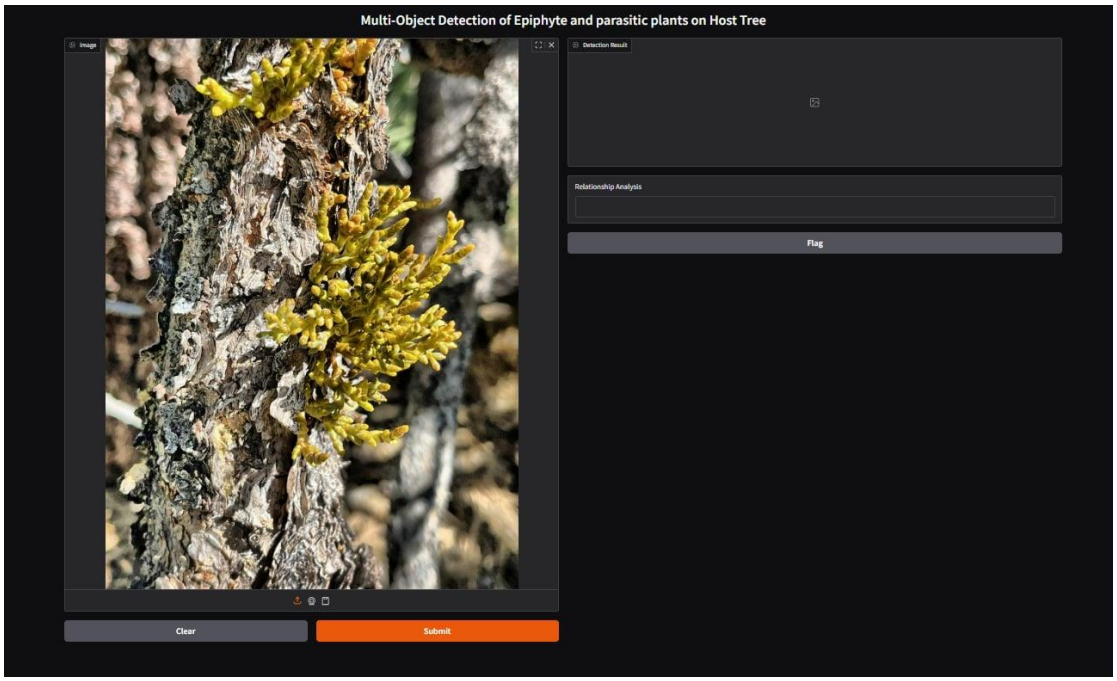
Date of Validation:

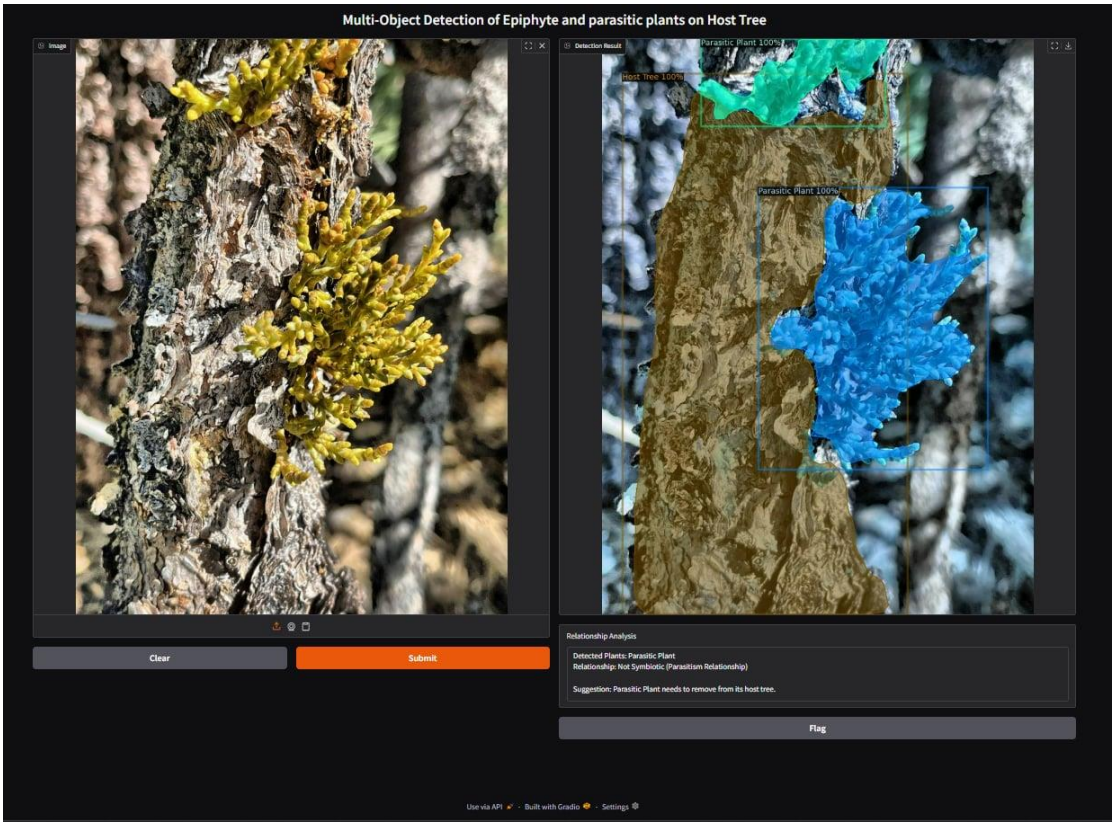
APPENDIX E

Sample Input / Output


Multi-Object Detection of Epiphyte and parasitic plants on Host Tree
image
Detection Result
Relationship Analysis
Flag
Clear
Submit
Use via API · Built with Gradio · Settings


Multi-Object Detection of Epiphyte and parasitic plants on Host Tree
Image
Detection Result
Relationship Analysis
Detected Plants: Epiphyte Plant
Relationship: Symbiotic (Commensalism Symbiotic Relationship)
Suggestion: Epiphyte Plant doesn't need to remove from its host tree.
Clear
Submit
Flag
Use via API · Built with Gradio · Settings


Multi-Object Detection of Epiphyte and parasitic plants on Host Tree
Image
Detection Result
Relationship Analysis
Flag
Clear
Submit


Multi-Object Detection of Epiphyte and parasitic plants on Host Tree
Image
Detection Result
Parasitic Plant 100%
Host Tree 100%
Parasitic Plant 100%
Clear
Submit
Relationship Analysis
Detected Plants: Parasitic Plant
Relationship: Not Symbiotic (Parasitism Relationship)
Suggestion: Parasitic Plant needs to remove from its host tree.
Flag
Use via API · Built with Gradio · Settings

APPENDIX F

Curriculum Vitae



CONTACT

0917-738-5012

rosalescarlosm19@gmail.com

Brgy. Bucal, Calamba City, Laguna Philippines, 4027

EDUCATION

Laguna State Polytechnic University - Los Baños
Bachelor of Science in Computer Science Major in Intelligent System
2022-2026

Calamba Institute
Science, Technology, Engineering, and Mathematics (STEM)
2020-2022

Calamba Institute
2016-2020

SKILLS

Video Editing
Time Management
Marketing
Teamwork
Communication
Adaptability

CARLOS MIGUEL Z. ROSALES

PROFILE

A hard-working, self-motivated, and reliable individual with experience in sales, customer service, and digital work. Works well both independently and as part of a team, demonstrating strong communication and time management skills. I can perform effectively under pressure and remain focused on achieving goals and completing tasks on time. Thrives on challenges to improve and always eager to learn new skills

EXPERIENCE

South Finds (2022 - 2024)

- Seller of thrifted clothes, bags, shoes, or caps either by operating in a thrift store or by reselling through online or elsewhere.
- Middle man between buyers and sellers of thrifts, helping them complete smooth and fair transactions. Handled communication, pricing, and coordination to ensure both sides were satisfied.

Bakery Store (2016 - 2025)

- Selling bread and other goods, and restocking products to ensure a consistent supply.
- Manages cash transactions accurately and maintains sales records.

LANGUAGE

Filipino
English



ALEJANDREI APOLO M. DURAN

CONTACT

0962-336-7401

alejandreiduran@gmail.com

11091, Tandang St., Mayondon, Los Baños, Laguna

EDUCATION

Laguna State Polytechnic University - Los Baños
Bachelor of Science in Computer Science Major in Intelligent System
2022-2026

Liceo De Los Baños
General Academic Strand (GAS)
2020-2022

SKILLS

Teamwork
Communication
Adaptability
Video Editing
Time Management

PROFILE

A motivated and reliable Computer Science student with hands-on experience in data encoding and basic web design. Comfortable using Microsoft Word and Excel for organizing and managing information. Skilled in creating simple web pages using HTML and CSS. Always eager to improve and learn new tools, especially in technology. I can work well on my own or with others, and I stay focused on finishing tasks correctly and on time.

EXPERIENCE

Data Encoder (Freelance / Work Experience)
(2022 - 2024)

- Entered and managed data using Microsoft Word and Excel
- Organized files and documents for easy tracking.
- Checked data for errors to make sure all information is correct.
- Helped support daily office tasks by creating simple reports and lists

Web Page Styling (HTML & CSS Projects)
(2022 - 2024)

- Created and edited basic web pages using HTML and CSS.
- Applied simple layouts, colors, fonts, and images to improve design
- Updated website sections to look clean and user-friendly.
- Learned how to organize website code for easy editing and updates.

LANGUAGE

Filipino
English

APPENDIX G

Matrix for Revisions

MATRIX FOR REVISIONS

Commentors	Consultants	Comments	Page No.	Actions	Page No.
Mr. Jefferson L. Lerrrios	Subject Specialist	Refine and improve the grammatical structure of the preliminary pages.	iii	The grammatical structure of the preliminary pages is refined and improved.	iii
Mr. Jefferson L. Lerrrios	Subject Specialist	Abstract must be 250 max words.	vi	Reduced the word counts from the abstract without losing the summary of the paper's substance	vi
Ms. Carolina R. Joval	Statistician	Improve the data from the table and explanation of the values in the table.	54 and 64	Removed the parentheses values in the tables and improved the explanation of the values in the tables.	53 and 62
Mr. Jhoco S. Millanes	Technical	Page Layout (Page Margins) and Table Format.	---	Each page of the paper is thoroughly checked and reviewed to ensure the right page margins and table formats.	--
Mr. Jhoco S. Millanes	Technical	Flow of the contents (interpretation - table number - table) and (interpretation – figure - figure number).	---	Flow of contents is carefully reviewed and revised to ensure proper sequencing of interpretation, table or figure number, and the corresponding table or figure.	--

APPENDIX H

Grammarly Report

Plant Symbiosis

by Grammarly

General metrics

80,720	11,557	571	46 min 13 sec	1 hr 28 min
characters	words	sentences	reading time	speaking time

Score



This text scores better than 95% of all texts checked by Grammarly

Writing Issues

152	2	150
Issues left	Critical	Advanced

Plagiarism

This text hasn't been checked for plagiarism

Plant Symbiosis

by Grammarly

General metrics

30,063	4,434	335	17 min 44 sec	34 min 6 sec
characters	words	sentences	reading time	speaking time

Score



This text scores better than 96% of all texts checked by Grammarly

Writing Issues

54	1	53
Issues left	Critical	Advanced

Plagiarism

This text hasn't been checked for plagiarism

APPENDIX I

Turnitin Report

V14_MANUSCRIPT.pdf

ORIGINALITY REPORT

13%	10%	12%	%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	www.mdpi.com Internet Source	1%
2	public-pages-files-2025.frontiersin.org Internet Source	1%
3	molescroftprimary.net Internet Source	1%
4	www.coursehero.com Internet Source	<1%
5	S.P. Jani, M. Adam Khan. "Applications of AI in Smart Technologies and Manufacturing", CRC Press, 2025 Publication	<1%
6	Ian Wasswa Sejinja, Neema Jakisa Owor, Seith N. Mugume. "A deep learning approach for urban pluvial flood depth estimation using citizen-generated data and physics-based validation", Journal of Hydroinformatics, 2025 Publication	<1%
7	Pushpa Choudhary, Sambit Satpathy, Arvind Dagur, Dhirendra Kumar Shukla. "Recent Trends in Intelligent Computing and Communication", CRC Press, 2025 Publication	<1%
8	kjei.edu.in Internet Source	<1%
9	www.frontiersin.org Internet Source	<1%

APPENDIX J

Publishable Paper (IEEE FORMAT)

DEEP LEARNING-BASED MULTI-OBJECT DETECTION OF EPIPHYTES AND PARASITIC PLANTS ON HOST TREES USING OPTIMIZED MASK R-CNN FOR SYMBIOTIC RELATIONSHIP

Carlos Miguel Z. Rosales
College of Computer Studies
Laguna State Polytechnic University – Los Baños Campus
Calamba City, Laguna, Philippines
rosalescarlosm19@gmail.com

Alejandrei Apolo M. Duran
College of Computer Studies
Laguna State Polytechnic University – Los Baños Campus
Los Baños, Laguna, Philippines
alejandreiduran@gmail.com

Crisanto F. Gulay
College of Computer Studies
Laguna State Polytechnic University – Los Baños Campus)
Los Baños, Laguna, Philippines
crisgulay@gmail.com

***Abstract*— The study aimed to implement and optimize a Mask R-CNN deep learning model to detect and segment epiphyte plants, parasitic plants, and host trees in a complex environment. Using 1661 augmented images, the model was trained with ResNet-50 and later improved with ResNet-101. The baseline achieved 73.088% mAP0.5 for bounding box and 67.694% for segmentation, while the optimized model reached 83.007% mAP0.5 and 76.746%, respectively. The model effectively detected small and partially occluded epiphyte plants and refined masks for parasitic plants and host trees. The overall implementation and optimization of the Mask R-CNN results suggest that the model can effectively detect and segment epiphyte plants, parasitic plants, and host trees for symbiotic relationships, making this baseline study more useful in forest biodiversity, forest health monitoring, and for future researchers for this field.**

Keywords*— *Mask R-CNN, epiphyte plants, parasitic plants, host trees, RestNet-50, RestNet-101, bounding box, segmentation, mAP0.5, symbiotic relationships, forest biodiversity, forest health monitoring

I. INTRODUCTION

Symbiosis refers to a long-term biological interaction between different species, which can be beneficial, neutral, or harmful to one or both partners [1]. In forest ecosystems, symbiotic relationships such as commensalism and parasitism play an important role in maintaining ecological balance and biodiversity [2]. Epiphyte plants, for instance, grow on trees without harming them, while parasitic plants depend on their hosts for nutrients, often affecting the host's growth [3]. Monitoring these organisms is important for assessing forest health; however, traditional methods are time-consuming and difficult to perform manually, often resulting in limited accuracy [4]. The lack of advanced monitoring tools and ecological datasets makes it difficult to track symbiotic interactions effectively [5].

To address these challenges, this study implements and optimized a Mask R-CNN model for detecting and segmenting epiphyte plants, parasitic plants, and host trees in natural outdoor images [6]. The model utilizes deep learning and computer vision techniques to automate identification, reducing manual effort while improving detection accuracy. By optimizing model parameters and training with augmented datasets, the model enhances detection and segmentation performance. This study supports biodiversity monitoring and conservation by introducing a practical approach to detect plant that is relevant to symbiotic relationships.

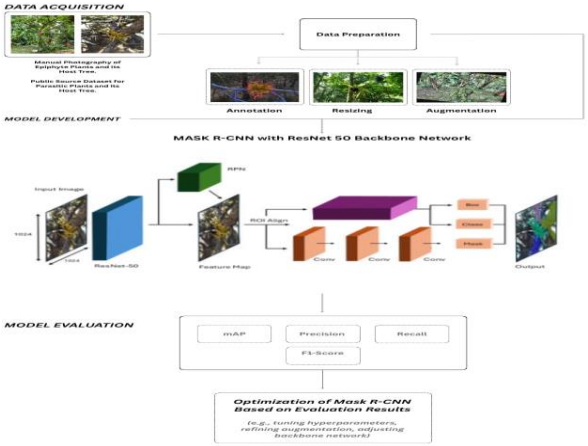
A. Objectives of the Study

The specific objectives of this study are:

1. To implement the Mask R-CNN model for multiple object detection and segmentation of epiphyte plants, parasitic plants, and host trees using still image data related to plant symbiosis.
2. To improve the detection and segmentation performance of a Mask R-CNN model for epiphyte plants and parasitic plants with their host trees by applying optimized data preparation and model tuning strategies.
3. To evaluate the performance of the Mask R-CNN model under real-world conditions using standard object detection and segmentation metrics within the study period.

B. Conceptual Framework

Figure 1. Conceptual Framework



XXX-X-XXXX-XXXX-X/XX/$XX.00 ©20XX IEEE

The study follows a structured framework consisting of three main stages: data acquisition, model development, and model evaluation. Image data of epiphyte and parasitic plants with their host trees were obtained through manual photography and supplemented with public datasets. Collected images were processed through annotation, resizing, and augmentation to improve data diversity and model performance.

The prepared dataset was used to train a Mask R-CNN model with a ResNet-50 backbone for object detection and instance segmentation. Model performance was evaluated using mAP, precision, recall, and F1-score. Based on the evaluation results, optimization was performed by tuning hyperparameters, refining augmentation techniques, and adjusting the backbone network to enhance accuracy and reliability.

II. METHODOLOGY

This section presents the development process of the proposed study consisting of data acquisition, annotation, preprocessing, augmentation, and splitting, model training, model evaluation, and model optimization

A. Data Acquisition

The dataset used in this study consists of three class categories, which are epiphyte plants, parasitic plants, and host trees. Initially, 212 images of epiphytes with host tree were collected through manual photography in Makiling Botanic Garden, and 344 images of parasitic plants with host tree were collected in a publicly available dataset. These methods were performed due to the limited availability of specific parasitic plants, so public datasets are relevant to the study. Images reflect real-world conditions such as natural lighting, varied plant structures, and organic growth patterns. Since the dataset is collected both manually and from public sources, there is a possibility of class imbalance, and to address this, additional augmentation will be applied selectively to underrepresented classes during preprocessing.

B. Annotation, Preprocessing, Augmentation and Splitting

Image annotation was performed using Roboflow, ensuring accurate labeling of epiphyte plants, parasitic plants, and host trees through polygon masks suitable for instance segmentation. This method provided precise pixel-level boundaries that allowed the Mask R-CNN model to effectively learn object shapes and distinctions.

For preprocessing, all images were resized to 1024×1024 pixels to maintain consistent input dimensions. To enhance dataset diversity and improve model robustness, data augmentation was applied, which included horizontal flipping, rotation ranging from −5° to +5°, brightness adjustment between −10% and +10%, and exposure adjustment between −5% and +5%. Additional transformations such as saturation, blur, and noise adjustments were also implemented to simulate various lighting and environmental conditions.

After preprocessing, the dataset was divided into 70% for training (1467 images), 15% for validation (97 images), and 15% for testing (97 images), resulting in a total of 1661 images with 7896 annotated instances across all classes. This structured preparation is crucial to ensure that the deep learning model can accurately learn and identify the relevant objects in each image, thereby improving its accuracy and robustness.

C. Model Training

In this study, the selected deep learning architecture is Mask R-CNN, initially implemented with a ResNet-50 backbone and default hyperparameters, which serves as a baseline for learning object features. If the model's performance during evaluation indicates challenges in object detection and segmentation, a deeper backbone architecture such as ResNet-101 together with some various adjusted hyperparameters will be introduced in the optimization phase to enhance feature representation and overall detection performance. The model will be implemented using the Detectron2 framework in Python, which supports instance segmentation, flexible backbone integration, and model hyperparameter configuration.

Since Mask R-CNN is capable of instance segmentation, it is expected to provide precise object boundaries, an essential feature for detecting plant structures that are often irregular, overlapping, or partially obscured. After initial training, the model's output will be reviewed to observe potential detection challenges or segmentation inconsistencies. These findings will guide further improvements in the model, which may include tuning hyperparameters such as learning rate or batch size, optimizing data preprocessing strategies, and adjusting the model configuration.

D. Model Evaluation

Once the model is trained, its performance is evaluated using standard object detection and segmentation metrics to understand how well it can detect, segment, and classify each object. These include mean Average Precision (mAP), precision, recall, F1-score, Intersection over Union (IoU), and the confusion matrix. These metrics help measure the model's ability to correctly detect objects, such as true positives, false positives or false negatives.

Evaluation is conducted on a separate set of images that were not included in the training process to test how well the model generalizes to new and unseen data. If the model fails to meet acceptable benchmarks, for example, if the mAP is below 0.5 or frequent confusion between classes, optimization techniques will be initiated. This step helps confirm the model's effectiveness and reliability in identifying plant symbionts and host trees in real-world conditions.

E. Model Optimization

After the initial training and testing phases, the model's performance will be reviewed to identify any potential limitations, such as detection inaccuracies, segmentation errors, or confusion between classes. Based on these observations, the model may undergo further optimization to enhance its overall performance.

One of the first steps in this process will be evaluating the backbone network. Initially, the model will be implemented using ResNet-50, but if performance issues occur, such as poor feature extraction or missed detections, a shift to a deeper backbone like ResNet-101 may be considered to improve accuracy and representation learning.

Other optimization efforts may involve refining data preprocessing steps, improving data augmentation strategies, or revisiting model configuration settings. These iterative adjustments are intended to improve the model's robustness

and accuracy, especially when dealing with the natural variability and visual complexity found in outdoor forest environments.

III. RESULTS AND DISCUSSION

The Implementation of Mask R-CNN for Multiple Object Detection and Segmentation of Epiphyte Plants, Parasitic Plants, and Host Trees

This section presents the initial implementation of the Mask R-CNN model using the original dataset to perform multiple object detection and segmentation of epiphyte plants, parasitic plants, and their host trees. The purpose of this phase is to establish the model's baseline performance before any optimization strategies are applied. It describes the training setup used, including the model configuration and hyperparameters, followed by the results obtained from the initial experiments. These results serve as the foundation for identifying the challenges and limitations of the model and the areas that require optimization in the succeeding section.

The Initial Training Setup of Mask R-CNN

This subsection explains the training setup used for the initial implementation of the Mask R-CNN model. It includes the dataset configuration, preprocessing procedures, and the hyperparameters applied during training. The goal of this setup is to ensure that the model is properly trained under consistent and controlled conditions, allowing for a fair evaluation of its baseline performance before optimization.

Table 1. Total Number of Annotated, Augmented, and Preprocessed Initial Image Datasets

Split	Number of Images
Training	1467
Validation	97
Test	97
Total	1661

Table 1 presents the total number of datasets after applying preprocessing and augmentations to the raw datasets. The 453 is the total number of the raw dataset, and the 1014 is the total of the augmented dataset from the training split. The overall raw dataset is split into 70/15/15, and the validation and test splits are not affected by the preprocessing and augmentations.

Table 2. Details of Preprocessing Used in Initial Model Training.

Preprocessing	Specification
Auto-Orient	Applied

Table 2 presents the preprocessing used for initial implementations. We applied Auto-Orient only and did not include resizing, as we used the default resizing from the Detectron2 framework.

Table 3. Details of Augmentations Used in Initial Model Training

Augmentations	Specifications
Flip	Horizontal
Rotation	Between -5° and +5°
Brightness	Between -10% and +10%
Exposure	Between -5% and +5%

Table 4. Details of Hyperparameters Used in Initial Model Training

Hyperparameter	Specifications
Learning rate	(0.02, 0.01, 0.005, 0.0025, 0.00125)
Weight Decay	0.0001
Batch Size	2
Epochs	10, 15, 20
Warmup Iterations	200
Number of Workers	2

The Result of the Initial Implementation of Mask R-CNN

The initial implementation of the Mask R-CNN for multiple object detection and segmentation of epiphyte plants, parasitic plants, and host trees has achieved quite low performance in terms of the object detection and segmentation metrics. These results suggest an opportunity for improvement and optimizations that we were preparing for. The results from this are also significant as they would be the baseline for this study.

Table 5 presents all the results of mAP50 from various hyperparameters. First, we trained the model with its default hyperparameters, such as a learning rate of 0.02 and the backbone architecture ResNet-50, and the rest of the configurations. However, during the model training with the default 0.02 learning rate, it crashes out. We then proceed with training with four more learning rates to see which learning rate would get the best results from the combinations and the deeper backbone network ResNet-101. One of the goals of proposing different learning rates and model configurations is to spot which area of the hyperparameters is best and well-suited to this dataset.

Table 5. Results from the Initial Implementation in Mask R-CNN

Learning Rate	Epochs	RestNet-50		RestNet-101	
		Bbox mAP50	Segm mAP50	Bbox mAP50	Segm mAP50
0.01	10	73.379	67.307	73.088	67.694
	15	71.987	67.790	69.733	65.789
	20	68.243	64.033	69.534	65.970
0.005	10	71.893	65.973	72.063	66.389
	15	69.916	64.591	70.465	66.671
	20	70.554	66.311	69.548	63.749
0.0025	10	71.000	65.324	69.200	63.607
	15	70.163	64.406	69.948	64.713
	20	70.269	65.326	69.965	64.646
0.00125	10	70.755	64.310	71.225	64.320
	15	70.742	64.910	68.139	62.880
	20	69.384	63.102	66.407	61.657

Table 6. Results of the Best Performing Initial Implementation of Mask R-CNN

Metrics	Bounding Box Level	Segmentation Level
mAP	50.019	34.714
mAP50	73.088	67.694
Precision	0.6839	0.6185
Recall	0.5934	0.5351
F1-Score	0.6346	0.5730

Table 6 shows the results of the best performing initial implementation of the Mask R-CNN model. At the bounding box level, the model obtained a mAP score of 50.019 and an mAP50 score of 73.088. For segmentation, the model achieved a mAP of 34.714 and an mAP50 of 67.694. The precision, recall, and F1-score metrics were evaluated at the optimal confidence threshold of 0.8 and at IoU50. The bounding box results show a precision of 0.6839, recall of 0.5934, and F1-score of 0.6346. Meanwhile, the segmentation level achieved a precision of 0.6185, recall of 0.5351, and F1-score of 0.5730. These values represent the performance of the model during its initial implementation stage.

After the initial implementation of the Mask R-CNN model, the results showed that modifying the hyperparameters and configurations gives the model better result metrics. Based on the table results, the default backbone architecture ResNet-50, achieved its best results in learning 0.01 and was trained on 10 epochs. Furthermore, by modifying its backbone architecture to ResNet-101, which is deeper than ResNet-50, it also showed an improvement for bounding box mAP50. Therefore, the initial implementation achieved its best results by modifying the backbone network from ResNet-50 to ResNet-101 with a learning rate of 0.01 in 10 10-epoch setup. This study proves that adjusting or modifying the hyperparameters improves its overall performance metrics.

However, based on the results table, when adjusting its epochs for each setup, it showed a decrease over time. So, in this initial implementation, 10 epochs is a well-suited duration for training the model, with a stable decrease in training loss and an increase in mAP every epoch. While adjusting the epochs, each setup showed a decrease, which also proves that the model overfits for longer training. Therefore, for the next phase of this study, it is suggested that optimizing and adjusting the data preparation and some model configurations would be better for more experimentation for further improvements of the model.

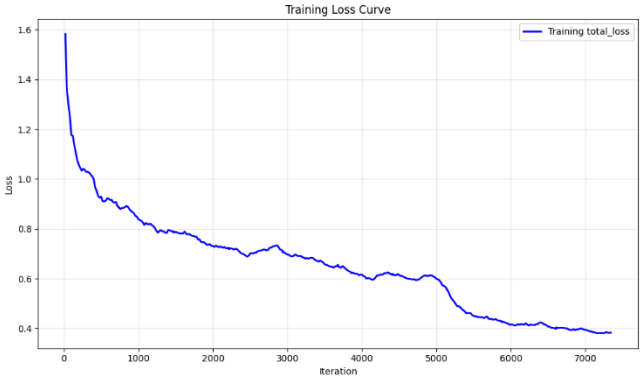


Figure 2. Training Loss of Best Result in Initial Implementation

Figure 2 presents the continuous decrease of training loss over the 10-epoch training, which suggests that the model is learning and not overfitting over the entire training process. Even with a slight increase in some of the iterations, which is an indicator of overfitting, it still found its way to learn and decrease the loss until the last epoch.

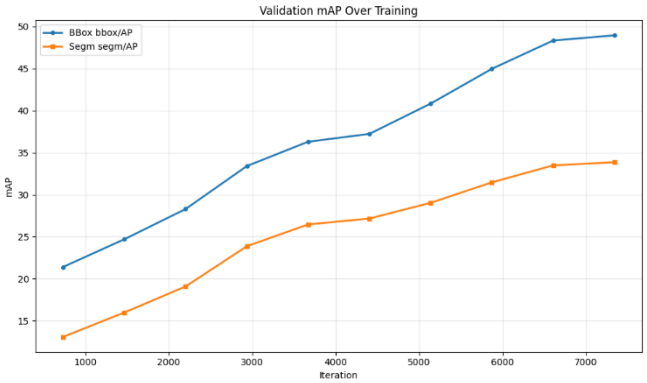


Figure 3. Validation mAP plot of Best Result in Initial Implementation

Figure 3 presents the plot of validation mAP over the entire training process. Based on the plot result, it smoothly increases over each epoch, which is an indicator that the model is learning.

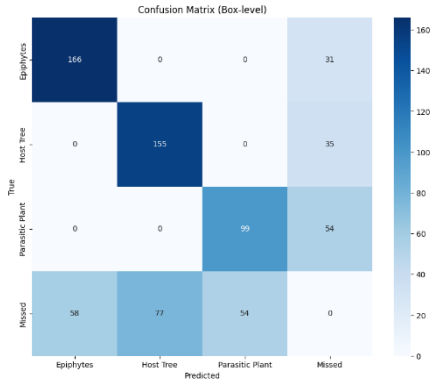


Figure 4. Confusion Matrix for bounding box level of Best

Result in Initial Implementation of Mask R-CNN

Figure 4 presents the confusion matrix for the bounding box level over the entire classes. The missed column represents the missed detection with each of the classes, and based on the table, it shows a minimal missed for each class. The missed row represents all the total confusion between each class, and based on the table, it just shows minimal confusion between each class.

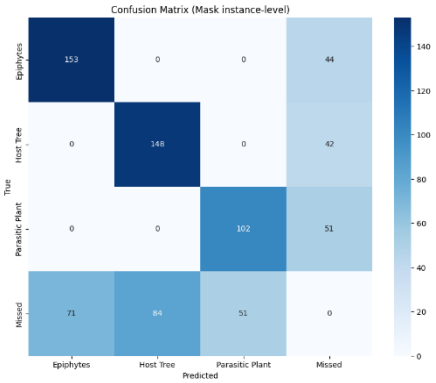


Figure 5. Confusion Matrix for bounding box level of Best Result in Initial Implementation of Mask R-CNN

Figure 5 presents the confusion matrix for the segmentation level over the entire classes. The missed column represents the missed segmentation with each of the classes, and the missed row is for confusion between each class. The table results show quite misses and confusion compared to the bounding box level, as the segmentation level is way more difficult task.

Performance Overview and Identification of Challenges

In this part, it overviews the model's performance after the initial implementation. Further observations are being carried out carefully to identify the challenges and weaknesses of the model in terms of detection and segmentation of the objects. Identified challenges would be significant for guiding the next phase of optimization of the model.



Figure 6. Overview of Object Detection and Segmentation of the Initial Output of Epiphyte Plants on Host Trees

Figure 6 presents the performance of the initial implementation of Mask R-CNN for detecting and segmenting the epiphyte plants and the host tree. Based on the result, it correctly detects and segments the epiphyte plants and their host tree.

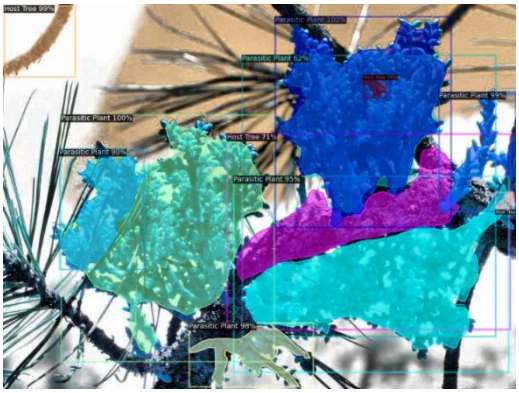


Figure 7. Overview of Object Detection and Segmentation of the Initial Output of Parasitic Plants on Host Trees

Figure 7 presents the performance of the initial implementation of Mask R-CNN for detecting and segmenting the parasitic plants and the host tree. Based on the result, it correctly detects and segments the parasitic plants and their host tree, even with some partially occluded objects.

After a careful and strict review of the model's performance under the test set dataset, some of the potential challenges are identified. For epiphyte plants on host trees, the model missed some of the small and partially occluded plants in detection and segmentation. Some of the epiphyte plants also have a poor or are not fully covered by the mask because of the too complex angle of the plant. While in parasitic plants on host trees, it also missed some of the parasitic plants and their host trees. The detection and segmentation position is also too complicated, as it doubles the predicted bounding box and mask with the target object. These identified challenges suggest further optimization of the training dataset so that the detection and segmentation for each epiphyte plant, parasitic plant, and host trees would be cleaner and not too complicated.

The Optimization of Mask R-CNN for Multiple Object Detection and Segmentation of Epiphyte Plants, Parasitic Plants, and Host Trees

The initial implementation of the Mask R-CNN model achieved great metrics and performance for the detection and segmentation of epiphyte plants, parasitic plants, and host trees. That initial implementation is highly significant as it would be the baseline for optimizing the model's metrics and performance based on the observed initial result. Further optimization strategies would be applied based on the observed problems and challenges after the implementation.

The Training Setup for Optimization of Mask R-CNN

The first step of optimization is removing some of the image datasets that have too many occluded plants and a complex background. Reducing the annotations within each image has also been applied for a cleaner output of detection and segmentation within an object. Furthermore, reviewing the number of instances per class has also been carefully monitored and balanced to avoid biases for each class.

The continuation of the model optimization process has added various preprocessing and augmentation techniques for further improvement of the model. For preprocessing, resizing to 1024x1024 is added to avoid cutting or removing some of the significant objects within image datasets, such as epiphyte plants, parasitic plants, and host trees. Auto-Adjust Contrast with a value of Adaptive Equalization in Roboflow is also applied to automatically improve and balance how bright and dark the areas of an image and for clearer details of the objects. Augmentations are also extended, such as rotation, saturation, brightness, exposure, blur, and noise, to improve the robustness of the model's performance in terms of natural and complex environments.

Table 7. Total Number of Cleaned and Optimized Datasets

Split	Number of Images
Training	1335
Validation	56
Test	55
Total	1446

Table 8. Total Number of Instances per Class of Cleaned and Optimized Datasets

Class	Number of Instances
Epiphyte Plant	1868
Parasitic Plant	1556
Host Tree	2310
Total	5734

Table 9. Details of Preprocessing Used in Optimized Model Training.

Preprocessing	Specification
Auto-Orient	Applied
Resizing	1024x1024
Auto-Adjust Contrast	Using Adaptive Equalization

Table 10. Details of Augmentations Used in Optimized Model Training

Augmentations	Specifications
Flip	Horizontal
Rotation	Between -15° and +15°
Brightness	Between -25% and +25%
Exposure	Between -5% and +5%
Saturation	Between -25% and +25%
Blur	Up to 2px
Noise	Up to 1% of pixels

Table 11. Details of Hyperparameters Used in Optimized Model Training

Hyperparameter	Specifications
Learning rate	(0.01, 0.005, 0.0025, 0.00125)
Weight Decay	0.0001
Batch Size	2
Epochs	10, 20, 30
Warmup Iterations	200
Number of Workers	2

The Result of the Implementation of Optimized Mask R-CNN

The implementation and application of the various techniques for optimization of the model's metrics and performance have achieved great improvement in mAP50 for bounding box and segmentation. The hyperparameters used from the initial implementation are retained except for epochs and the learning 0.02, as it crashes out during the initial implementation. For this optimization, we trained a little longer for 10, 20, and 30 epochs to give the model a chance to learn more, and the rest of the hyperparameters and model configurations are retained.

Table 12. Results from the Optimized Implementation in Mask R-CNN

Learning Rate	Epochs	RestNet-50		RestNet-101	
		Bbox mAP50	Segm mAP50	Bbox mAP50	Segm mAP50
0.01	10	83.482	75.104	82.783	75.064
	20	81.709	76.490	82.182	76.808
	30	80.019	76.221	79.220	74.491
0.005	10	82.666	74.080	83.007	76.746
	20	78.986	72.461	77.905	72.232
	30	78.784	73.388	79.065	73.563
0.0025	10	83.802	71.742	82.158	72.903
	20	79.938	69.811	81.015	75.396
	30	79.110	71.627	79.260	72.389
0.00125	10	82.624	69.682	82.241	71.392
	20	80.143	70.041	81.537	71.372
	30	79.462	68.474	79.946	72.665

Table 12 presents the optimized Mask R-CNN, highlighting the significant impact of how the various optimization strategies greatly affect the improvement of the mAP50. After a careful evaluation of each result, the best and balanced results for bounding box and segmentation were achieved at a learning rate of 0.005 in 10 epochs with a backbone network of ResNet-101. This demonstrates that the optimization of hyperparameters and the refinement of data preparation significantly enhance the overall metrics and performance of the model.

Table 13. Results of the Best Performing Optimized Implementation of Mask R-CNN

Metrics	Bounding Box Level	Segmentation Level
mAP	58.811	37.082
mAP50	83.007	76.746
Precision	0.6693	0.5688
Recall	0.6849	0.5820
F1-Score	0.6768	0.5751

Table 13 presents the performance results of the best optimized Mask R-CNN model. For the bounding box level, the model achieved a mAP score of 58.811 and an mAP50 score of 83.007. At the segmentation level, the model recorded a mAP of 37.082 and an mAP50 of 76.746. The precision, recall, and F1-score values were also evaluated at the optimal confidence threshold of 0.8 and at an IoU50 threshold. The bounding box level reached a precision of 0.6693, recall of 0.6849, and F1-score of 0.6768. For segmentation, the precision was 0.5688, recall was 0.5820, and the F1-score was 0.5751. These values represent the final performance of the optimized model used in this study.

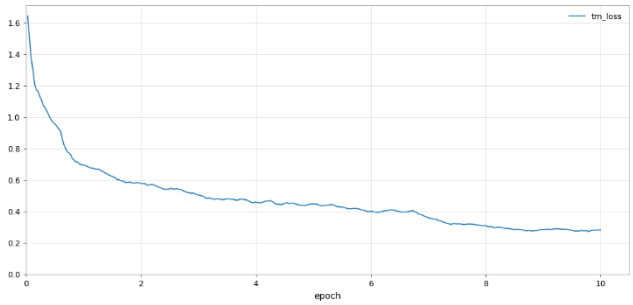


Figure 8. Training Loss Plot of Best Result in Optimized Mask R-CNN

Figure 8 presents the smooth decrease of training loss for the optimized Mask R-CNN, which indicates that the model is smoothly learning with the optimization techniques used. Even with a minimal increase in some of the iterations, the model found its way to decrease the loss and learn more.

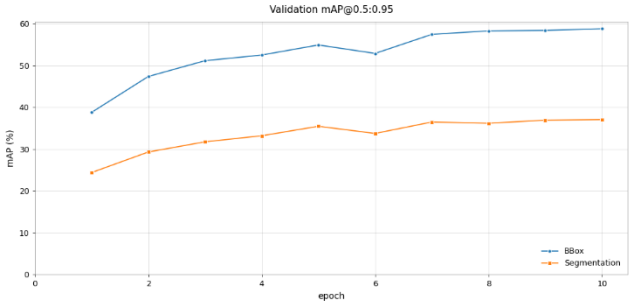


Figure 9. Validation mAP Plot of Best Result in Optimized Mask R-CNN

Figure 9 presents the validation mAP of the optimized Mask R-CNN over the entire epochs. Based on the plot results, although the plot shows a very light increase each epoch, it doesn't suggest that the model fluctuates or overfits. A light or very low increase over time is still progress for the model.

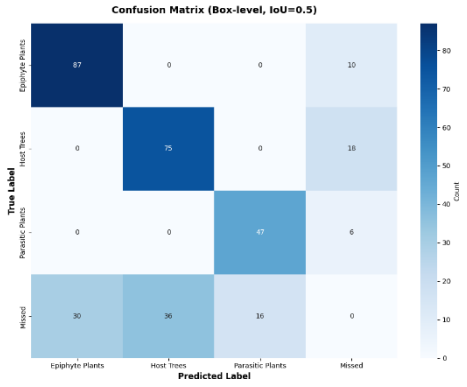


Figure 10. Confusion Matrix for bounding box level of Best Result in Optimized Mask R-CNN

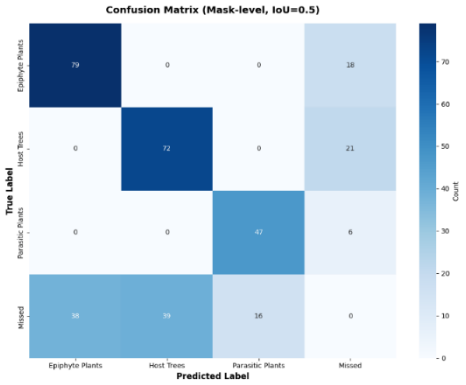


Figure 11. Confusion Matrix for the segmentation level of Best Result in Optimized Mask R-CNN

Figure 10 and Figure 11 presents the confusion matrix for both the bounding box and segmentation level for optimized Mask R-CNN. Based on the table result, both show a great detection and segmentation for each class, and minimal missed objects and confusion for each class for the bounding box and segmentation level compared to the initial implementation.

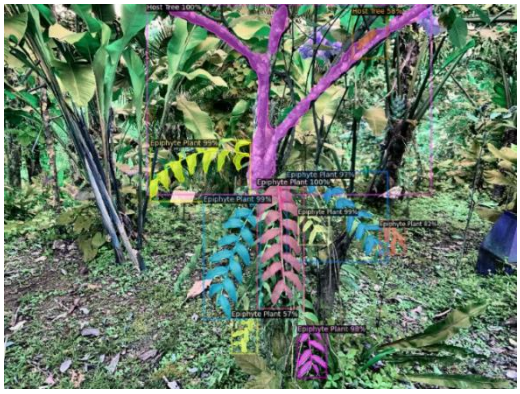


Figure 12. Overview of Object Detection and Segmentation of Optimized Output of Epiphyte Plants on Host Trees

Figure 12 presents the performance of the optimized Mask R-CNN based on the various optimization techniques applied. Based on the performance of the model in epiphyte plants with their host tree, it successfully detects and segments, and even refines the mask over the target objects, especially the small objects that are not present in the initial implementation.

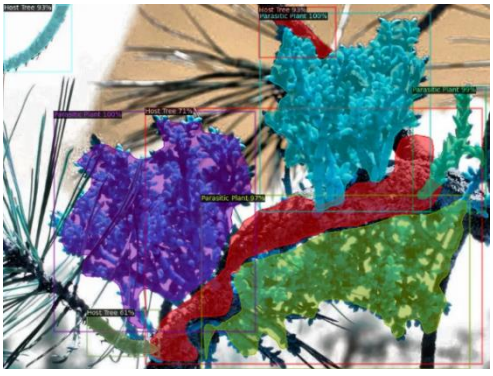


Figure 13. Overview of Object Detection and Segmentation of Optimized Output of Epiphyte Plants on Host Trees

Figure 13 presents the performance of the optimized Mask R-CNN on parasitic plants and their host tree. Based on the results, it is also successful in terms of detection and segmentation, and even refines the mask over the parasitic plants. Based on the observations, compared to the initial implementation result, the mask is now cleaner and clearer, which means it doesn't produce excess mask over the objects.

Statistical Test

Table 14. Descriptive Statistics for Both Expert Evaluation and Optimized Mask R-CNN Prediction

Group	N	Mean	Median	SD	SE
Botanist Expert	100	1.50	1.5	0.503	0.050
Model	100	1.56	2.0	0.556	0.056

Table 14 presents the descriptive statistics of the Expert and Model scores based on one hundred (100) image samples. This presents the detailed comparison, highlighting any significant differences or alignments between the expert's evaluations and the model's predictions. This analysis is crucial for validating the effectiveness of the Mask R-CNN model in detecting and segmenting epiphyte/parasitic plants with their host tree and ensuring its applicability and practicality.

The expert obtained a mean score of 1.50 and a median of 1.5, indicating that the majority of expert evaluation were centered between the score of 1 and 2. The standard deviation (SD = 0.503) and standard error (SE = 0.050) suggest minimal variability among the expert evaluations, implying consistent scoring across the samples.

Meanwhile, the model achieved a slightly higher mean score of 1.56 with a median of 2.0, suggesting that the model tends to predict slightly a scores of 3 compared to the expert. The standard deviation (SD = 0.556) and standard error (SE = 0.056) indicate low variability, reflecting that the model's outputs are consistent and closely aligned with the expert's evaluations.

Table 15. Result of Paired Samples T-Test for Both Expert Evaluation and Optimized Mask R-CNN Prediction

T-Test	Statistic	Df	P	Mean Difference	Effect Size (Cohen's d)
Paired T-test	-1.75	99	0.083	-0.06	-0.175

Table 15 presents paired sample t-test revealed a *t*-value of -1.75 with 99 degrees of freedom, indicating a small difference between the expert evaluation and model prediction. The computed *p*-value of 0.083, which is greater than the 0.05 significance threshold, indicates that there is no significant difference. This means that the optimized Mask R-CNN model performs comparably to experts in identifying these plant types within the dataset. The mean difference of -0.06 further supports this, showing that the model's prediction scores are only slightly higher than the expert evaluations. Moreover, the computed effect size (Cohen's *d* = -0.175) indicates a small effect, emphasizing that the variation between the two sets of scores is minimal. These results demonstrate that the deep learning-based model effectively reflects expert-level performance, confirming its reliability in detection and segmentation of epiphytes or parasitic plants with their host tree.

IV. CONCLUSION

1. **To implement the Mask R-CNN model for multiple object detection and segmentation of epiphyte plants, parasitic plants, and host trees using still image data related to plant symbiosis.**

   This study successfully implements the deep learning-based Mask R-CNN, capable of detecting and segmenting epiphyte plants, parasitic plants, and host trees for symbiotic relationship. From data preparation to hyperparameter tuning and adjusting model configuration, the model achieved 74.598% of mAP50 at the bounding box level and 68.740% mAP50 at the segmentation level, with a ResNet-101 backbone architecture for initial implementation and baseline of the study. This study proves that manually tuning the hyperparameter and configuration has a significant improvement in standard metrics.

2. **To improve the detection and segmentation performance of a Mask R-CNN model for epiphyte plants and parasitic plants with their host trees by applying optimized data preparation and model tuning strategies.**

   The application of various optimization strategies, from data cleaning, adding proper preprocessing and augmentations, hyperparameter tuning, and model configuration, the model achieved a huge improvement, from 74.598% to 83.007% of mAP50 at the bounding box level and from 68.740% to 76.746% of mAP50 at the segmentation level. This suggests that by carefully adjusting and optimizing the data preparations, hyperparameters, and model configurations, the model was successfully optimized with it.

3. **To evaluate the performance of the Mask R-CNN model under real-world conditions using standard object detection and segmentation metrics within the study period.**

   The performance of the Mask R-CNN model was successfully tested using real-world image data to see how well it can detect and segment epiphyte plants, parasitic plants, and host trees in natural environments. The model showed consistent and reliable results, maintaining high performance accuracy even with varying lighting, backgrounds, and plant positions. The use of standard evaluation metrics such as mAP50 for both detection and segmentation confirmed that the model performs effectively beyond controlled testing conditions. This signifies the Mask R-CNN model can be applied in actual field scenarios to help in identifying and analyzing plant symbiotic relationships.

ACKNOWLEDGMENT

The authors express their sincere gratitude to Almighty God for guidance and strength throughout this research. They also extend their appreciation to Laguna State Polytechnic University for providing a supportive academic environment. Special thanks are given to their research adviser and to the university administrators, faculty members, and research panel for their valuable guidance and feedback. The authors also acknowledge the Makiling Botanic Garden and its staff for their assistance and cooperation during data collection and validation.

REFERENCES

[1] R. P. Singh and A. Kumar, *Symbiosis and Plant Interactions*, Springer, 2025.

[2] M. A. Van Der Heijden, R. D. Bardgett, and N. M. van Straalen, "The unseen majority: soil microbes as drivers of plant diversity and productivity," *Ecology Letters*, vol. 8, no. 9, pp. 993–1001, 2015.

[3] G. Zotz, "The systematic distribution of vascular epiphytes—a critical update," *Botanical Journal of the Linnean Society*, vol. 171, no. 3, pp. 453–481, 2013.

[4] M. Zhao, X. Liu, and J. Wang, "Challenges of manual forest biodiversity monitoring and the rise of automated detection," *Ecological Informatics*, vol. 52, pp. 23–31, 2019.

[5] E. Ramos and P. Malabrigo, "Biodiversity monitoring and conservation gaps in Philippine forests," *Philippine Journal of Science*, vol. 149, no. 1, pp. 45–58, 2020.

[6] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 2961–2969.